

International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Lab 3 - Data Carving with XXD, Binwalk and Scalpel

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EXECUTIVE SUMMARY

This practical laboratory exercise focused on the identification, recovery, and analysis of digital evidence using file carving and forensic analysis techniques. The investigation involved a JPEG image and a FAT12 forensic disk image containing deleted files.

The JPEG evidence was examined at the binary level using xxd, analyzed for metadata using ExifTool, and examined with strings and Binwalk. The FAT12 forensic image was analyzed using The Sleuth Kit to identify filesystem structures and deleted files. Deleted files were subsequently recovered using icat and independently examined using Scalpel to demonstrate signature-based file carving.

The investigation successfully recovered four deleted files from the FAT12 image: two Microsoft Word documents and two text files. Hashing was used to preserve and verify evidence integrity. Scalpel independently identified Word-document content but also produced duplicate, incomplete, and over-carved results, demonstrating the limitations of signature-based carving compared with filesystem-aware recovery.

Overall, the exercise demonstrated practical digital forensic principles including evidence preservation, filesystem analysis, deleted-file recovery, file carving, metadata examination, hashing, and validation of forensic findings.

PURPOSE OF THE ASSIGNMENT

The purpose of this assignment was to develop practical skills in the examination and recovery of digital evidence using forensic tools and file-carving techniques.

The exercise provided hands-on experience in examining files at the binary level, identifying file structures and metadata, analyzing a forensic disk image, recovering deleted files, and comparing different forensic recovery techniques.

The assignment also aimed to demonstrate the importance of maintaining evidence integrity and validating forensic findings through independent analysis and cryptographic hashing.

OBJECTIVES OF THE ASSIGNMENT

The objectives of the assignment were to:

- Examine digital evidence using hexadecimal and binary-level analysis.
- Identify file headers, footers, and structural signatures.
- Analyze JPEG metadata and embedded information.
- Calculate and document cryptographic hashes for evidence integrity.
- Identify the filesystem structure of a forensic disk image.
- Analyze a FAT12 filesystem using The Sleuth Kit.
- Identify deleted files within a forensic image.
- Recover deleted files using filesystem-aware forensic techniques.
- Perform file carving using Scalpel.
- Compare filesystem-based recovery with signature-based carving.
- Identify valid recoveries, duplicates, incomplete files, false positives, and over-carved data.
- Validate recovered evidence using file identification, strings analysis, metadata, and cryptographic hashes.
- Document forensic findings in a clear and reproducible manner.

METHODOLOGY OF THE ASSIGNMENT

The investigation followed a structured forensic workflow designed to preserve evidence integrity and allow the results to be independently verified.

1. Evidence Preservation and Hashing

The original evidence files were stored separately from working copies and recovered files. MD5 and SHA-256 hashes were calculated for the original evidence to establish integrity and provide reference values for subsequent verification.

2. Binary-Level JPEG Examination

The JPEG evidence was examined using xxd to identify its hexadecimal structure, including the JPEG Start of Image (SOI) and End of Image (EOI) markers.

strings was used to identify readable text within the image, while ExifTool was used to extract metadata such as the camera manufacturer and model.

Binwalk was subsequently used to identify recognizable file signatures and embedded structures within the JPEG.

3. Forensic Disk Image Analysis

The Ch01InChap01.dd forensic image was examined using The Sleuth Kit.

img_stat was used to determine the image type and size. mmls was used to check for partition structures. Since no partition table was identified, the filesystem was analyzed from the beginning of the image.

fsstat identified the filesystem as FAT12 and provided information about the filesystem layout, including the FAT, root directory, cluster area, sector size, and cluster size.

4. Deleted File Identification and Recovery

fls was used to recursively examine the filesystem and identify deleted directory entries. Four deleted files were identified:

- Billing Letter.doc
- confirmation.txt
- letter1.txt
- Regrets.doc

The deleted files were recovered using icat and stored separately in the recovered directory.

5. Recovered File Examination

The recovered files were examined using file to determine their file types. strings was used to extract readable content from the recovered Word documents.

The recovered Word documents were also examined for embedded metadata, including document authorship, application information, timestamps, and document statistics.

6. Binwalk Analysis of the Forensic Image

Binwalk was executed against the FAT12 forensic image to search for additional recognizable embedded file signatures. No additional signatures were identified.

7. Scalpel File Carving

Scalpel was configured to recognize JPEG and Microsoft Word document signatures. It was then executed against the forensic disk image as an independent signature-based carving technique.

Scalpel produced ten DOC candidates. Several were incomplete or invalid, while others were identified as valid Word documents or duplicates. No JPEG files were carved.

8. Comparative Validation

The Scalpel results were compared with the files recovered using The Sleuth Kit. File types, document metadata, readable content, file sizes, and MD5/SHA-256 hashes were compared.

Although Scalpel recovered content corresponding to the deleted Word documents, its output files were not byte-for-byte identical to the Sleuth Kit recoveries. The results demonstrated the difference between filesystem-aware recovery and signature-based carving.

TOOLS AND RESOURCES USED

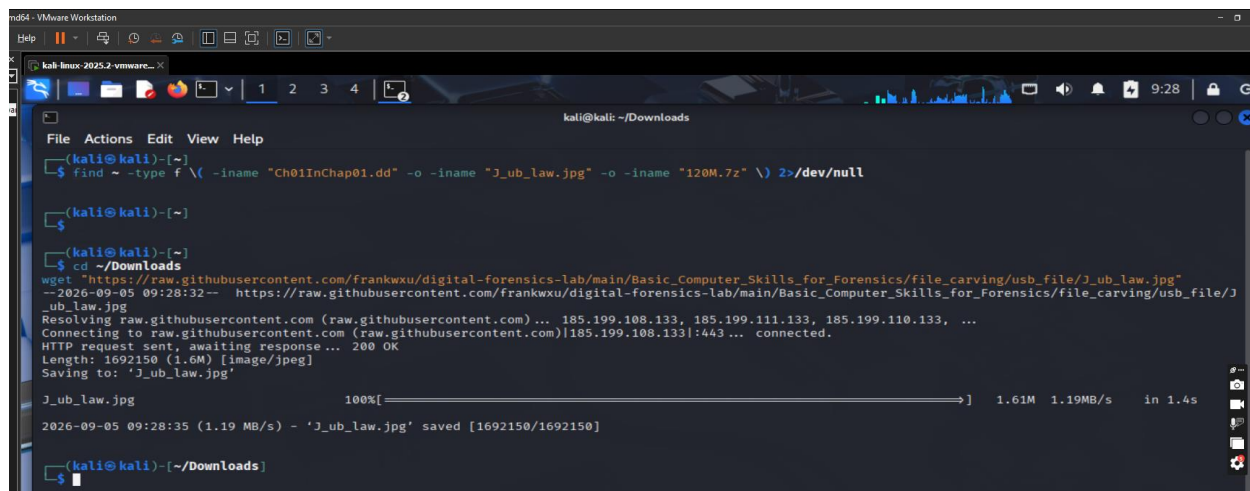
The following tools and resources were used during the investigation:

Tool / Resource	Purpose
Kali Linux	Digital forensic analysis environment
xxd	Hexadecimal/binary examination
strings	Extraction of readable strings from files
ExifTool	JPEG metadata extraction
Binwalk	File signature and embedded-data analysis
The Sleuth Kit	Filesystem and forensic image analysis
img_stat	Forensic image information
mmls	Partition/layout analysis
fsstat	Filesystem structure analysis
fls	File and deleted-entry identification
icat	Recovery of files from forensic images
Scalpel	Signature-based file carving
md5sum	MD5 hash generation and verification

sha256sum	SHA-256 hash generation and verification
GitHub	Documentation and version-controlled storage of the completed lab
Forensic evidence files	JPEG image, FAT12 disk image, and compressed evidence archive

LABS

Screenshot of the J_ub_law.jpg - JPEG Sample for Hex and Signature Analysis download



```

kali@kali: ~/Downloads
File Actions Edit View Help
(kali@kali)-[~]
$ find ~ -type f \( -iname "Ch01InChap01.dd" -o -iname "J_ub_law.jpg" -o -iname "120M.7z" \) 2>/dev/null

(kali@kali)-[~]
$
(kali@kali)-[~]
$ cd ~/Downloads
$ wget "https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_file/J_ub_law.jpg"
--2026-09-05 09:28:32-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_file/J_ub_law.jpg
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1692150 (1.6M) [image/jpeg]
Saving to: 'J_ub_law.jpg'

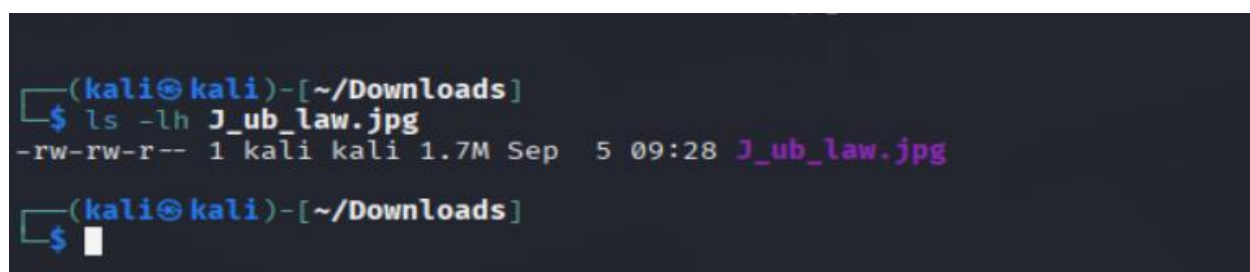
J_ub_law.jpg
100%[=====] 1.61M 1.19MB/s in 1.4s

2026-09-05 09:28:35 (1.19 MB/s) - 'J_ub_law.jpg' saved [1692150/1692150]

(kali@kali)-[~/Downloads]
$

```

Download verification



```

(kali@kali)-[~/Downloads]
$ ls -lh J_ub_law.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  5 09:28 J_ub_law.jpg

(kali@kali)-[~/Downloads]
$

```

Evidence of download and verification of 120M.7z — USB Image for Scalpel File Carving

```
(kali@kali)-[~/Downloads]
$ ls -lh J_ub_law.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  5 09:28 J_ub_law.jpg

(kali@kali)-[~/Downloads]
$ cd ~/Downloads
wget "https://github.com/frankwxu/digital-forensics-lab/raw/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z"
--2026-09-05 09:46:37-- https://github.com/frankwxu/digital-forensics-lab/raw/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z
Resolving github.com (github.com)... 140.82.121.4
Connecting to github.com (github.com)[140.82.121.4]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z [following]
--2026-09-05 09:46:38-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 36720470 (35M) [application/octet-stream]
Saving to: '120M.7z'

120M.7z          100%[=====] 35.02M  5.18MB/s  in 13s

2026-09-05 09:46:54 (2.79 MB/s) - '120M.7z' saved [36720470/36720470]

(kali@kali)-[~/Downloads]
$ ls -lh 120M.7z
-rw-rw-r-- 1 kali kali 36M Sep  5 09:46 120M.7z

(kali@kali)-[~/Downloads]
$
```

Evidence of downloading Ch01InChap01.dd — Forensic DD Image

```
(kali@kali)-[~/Lab3_Data_Carving]
$ cd ~/Downloads
wget -O Ch01InChap01.dd "https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd?dl=1"
--2026-09-05 09:55:53-- https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd?dl=1
Resolving www.dropbox.com (www.dropbox.com)... 162.125.64.18, 2620:100:6028:18::a27d:4712
Connecting to www.dropbox.com (www.dropbox.com)[162.125.64.18]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/scl/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mlxgdydc6wdwu0cro6dl=1 [following]
--2026-09-05 09:55:54-- https://www.dropbox.com/scl/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mlxgdydc6wdwu0cro6dl=1
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://ucd9d5793f2e901fc99cf9ec2ac9.dl.dropboxusercontent.com/cd/0/inline/DHjK3DvnR36HmN9AMu61LSqkM0oMw-fKjZEz-Fqns3VQF1U96Kd_LSfUMFBKemuiiK0fZEt6nziKBJPHKgNppPrw9gJcPwQkD0aPB2DFyJhG-1aGt1h_X2-63hCXq8dgDY/file?dl=1 [following]
--2026-09-05 09:55:56-- https://ucd9d5793f2e901fc99cf9ec2ac9.dl.dropboxusercontent.com/cd/0/inline/DHjK3DvnR36HmN9AMu61LSqkM0oMw-fKjZEz-Fqns3VQF1U96Kd_LSfUMFBKemuiiK0fZEt6nziKBJPHKgNppPrw9gJcPwQkD0aPB2DFyJhG-1aGt1h_X2-63hCXq8dgDY/file?dl=1
Resolving ucd9d5793f2e901fc99cf9ec2ac9.dl.dropboxusercontent.com (ucd9d5793f2e901fc99cf9ec2ac9.dl.dropboxusercontent.com)... 162.125.64.15, 2620:100:6028:15::a27d:470f
Connecting to ucd9d5793f2e901fc99cf9ec2ac9.dl.dropboxusercontent.com (ucd9d5793f2e901fc99cf9ec2ac9.dl.dropboxusercontent.com)[162.125.64.15]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DHj4MAkcqgsZEmMSco05aiZltefwIc6DhE2VoYfHwV4M_G0p-zJOGh4kN04ZmDa-U0luYGZTW2EAsv0-GW_ULZh06QvbJiQ5Xr0H5C8ymVvdGtnLFyFv95eSVGxk4R61NzTcL9CWTWCh0wYG28bqkq1w_68ZiWunqnVkBv-0Nv0JuIxr1wPZnLC3J1ZJpPtD2_Fx-15eUj6FDtm0ujsDnqThGlb3zz247fw9PHONvCRxawnh0ZTU_8KPL_Y0jv8UDFFwLZ5r1szZlbNn3Ziq0Q5PicsGhBg_vVnNCbnat_LPam9yp805DoDAK74TLw4GYivFg0FYtICfQ6ywbJ-bKHcZ9Z03k5PrY_2Eizrvmq/file?dl=1 [following]
--2026-09-05 09:55:59-- https://ucd9d5793f2e901fc99cf9ec2ac9.dl.dropboxusercontent.com/cd/0/inline2/DHj4MAkcqgsZEmMSco05aiZltefwIc6DhE2VoYfHwV4M_G0p-zJOGh4kN04ZmDa-U0luYGZTW2EAsv0-GW_ULZh06QvbJiQ5Xr0H5C8ymVvdGtnLFyFv95eSVGxk4R61NzTcL9CWTWCh0wYG28bqkq1w_68ZiWunqnVkBv-0Nv0JuIxr1wPZnLC3J1ZJpPtD2_Fx-15eUj6FDtm0ujsDnqThGlb3zz247fw9PHONvCRxawnh0ZTU_8KPL_Y0jv8UDFFwLZ5r1szZlbNn3Ziq0Q5PicsGhBg_vVnNCbnat_LPam9yp805DoDAK74TLw4GYivFg0FYtICfQ6ywbJ-bKHcZ9Z03k5PrY_2Eizrvmq/file?dl=1
Reusing existing connection to ucd9d5793f2e901fc99cf9ec2ac9.dl.dropboxusercontent.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 1474560 (1.4M) [application/binary]
Saving to: 'Ch01InChap01.dd'

Ch01InChap01.dd  100%[=====] 1.41M  180KB/s  in 24s

2026-09-05 09:56:24 (60.0 KB/s) - 'Ch01InChap01.dd' saved [1474560/1474560]

(kali@kali)-[~/Downloads]
$ ls -lh Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.5M Sep  5 09:56 Ch01InChap01.dd

(kali@kali)-[~/Downloads]
$
```

Evidence of copying all three into the evidence directory

```
(kali@kali)-[~/Downloads]
$ cp ~/Downloads/Ch01InChap01.dd ~/Lab3_Data_Carving/original_evidence/
cp ~/Downloads/J_ub_law.jpg ~/Lab3_Data_Carving/original_evidence/
cp ~/Downloads/120M.7z ~/Lab3_Data_Carving/original_evidence/

(kali@kali)-[~/Downloads]
$ ls -lh ~/Lab3_Data_Carving/original_evidence/
total 39M
-rw-rw-r-- 1 kali kali 36M Sep  5 09:58 120M.7z
-rw-rw-r-- 1 kali kali 1.5M Sep  5 09:58 Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.7M Sep  5 09:58 J_ub_law.jpg
```

MD5 Calculation of all downloads

```
(kali@kali)-[~/Downloads]
$ cd ~/Lab3_Data_Carving/original_evidence
md5sum * | tee ../hashes/original_md5.txt
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
```

SHA-256 calculation of all downloads

```
(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ sha256sum * | tee ../hashes/original_sha256.txt
2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce 120M.7z
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg
```

Complete display of all hashes

```
(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ cat ../hashes/original_md5.txt
cat ../hashes/original_sha256.txt
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce 120M.7z
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg
```

The initial forensic hash acquisition is complete and the outputs showed the baseline integrity values:

Evidence	MD5	SHA-256
120M.7z	dfe7b5424e54cd1bf50d5df47aceeb3c	2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce
Ch01InChap01.dd	a117773bcf1fc88ec0ab8e0a349fbbcb	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122
J_ub_law.jpg	83a360ac7f7e0ca318e5bfe39f95f137	238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468

Part A: JPEG Hex Analysis

A working copy of the supplied J_ub_law.jpg evidence file was created in the forensic working directory to ensure that the original evidence remained preserved and unmodified. The file utility was first used to confirm the file type and identify available

image information. The output identified the file as JPEG image data conforming to the EXIF standard. It also identified an embedded TIFF/EXIF structure and reported the camera manufacturer as NIKON CORPORATION and the camera model as NIKON D4. The image was reported as having dimensions of 1920 × 1080 pixels.

```
(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ cp ~/Lab3_Data_Carving/original_evidence/J_ub_law.jpg ~/Lab3_Data_Carving/working/J_ub_law_working.jpg
(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ cd ~/Lab3_Data_Carving/working
(kali@kali)-[~/Lab3_Data_Carving/working]
$ file J_ub_law_working.jpg
J_ub_law_working.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, dentries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
(kali@kali)-[~/Lab3_Data_Carving/working]
$ ls -lh J_ub_law_working.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  5 10:04 J_ub_law_working.jpg
```

Direct hexadecimal examination was then performed using xxd. Examination of the first 32 bytes of the file produced the sequence FF D8 FF E1, with FF D8 representing the JPEG Start of Image (SOI) marker. The subsequent FF E1 marker indicates an APP1 segment, and the visible Exif string confirms the presence of EXIF-related data within the file.

```
(kali@kali)-[~/Lab3_Data_Carving/working]
$ xxd -l 32 J_ub_law_working.jpg
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ... .. Exif .. II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ... .. @.

(kali@kali)-[~/Lab3_Data_Carving/working]
$ xxd J_ub_law_working.jpg | tail -5
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80  )O.J.Y.. (.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45  ....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1  .*.-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565  ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9  .Q....
```

The end of the file was subsequently examined using xxd and the final five lines of the hexadecimal output were reviewed. The final bytes contained FF D9, which represents the JPEG End of Image (EOI) marker. The presence of both the expected SOI (FF D8) and EOI (FF D9) markers provides direct binary-level evidence supporting the identification of the file as a valid JPEG structure.

```
(kali@kali)-[~/Lab3_Data_Carving/working]
$ xxd -l 32 J_ub_law_working.jpg
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ... .. Exif .. II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ... .. @.

(kali@kali)-[~/Lab3_Data_Carving/working]
$ xxd J_ub_law_working.jpg | tail -5
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80  )O.J.Y.. (.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45  ....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1  .*.-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565  ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9  .Q....
```

The hexadecimal header and footer outputs were saved separately in the forensic analysis directory as jpeg_header.txt and jpeg_footer.txt. These outputs provide a repeatable record of the signature analysis and were retained as supporting evidence for the forensic report.

```
(kali@kali)-[~/Lab3_Data_Carving/working]
$ xxd -l 32 J_ub_law_working.jpg > ../xxd_analysis/jpeg_header.txt
xxd J_ub_law_working.jpg | tail -5 > ../xxd_analysis/jpeg_footer.txt

(kali@kali)-[~/Lab3_Data_Carving/working]
$ cat ../xxd_analysis/jpeg_header.txt
cat ../xxd_analysis/jpeg_footer.txt
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80 )O.J.Y..(.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45 .....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1 *.-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.8..o..Z.T.e
```

02_xxd_jpeg_header_footer.png

```
(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ cd ~/Lab3_Data_Carving/working

(kali@kali)-[~/Lab3_Data_Carving/working]
$ file J_ub_law_working.jpg
J_ub_law_working.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpr
etation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3

(kali@kali)-[~/Lab3_Data_Carving/working]
$ ls -lh J_ub_law_working.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  5 10:04 J_ub_law_working.jpg

(kali@kali)-[~/Lab3_Data_Carving/working]
$ xxd -l 32 J_ub_law_working.jpg
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.

(kali@kali)-[~/Lab3_Data_Carving/working]
$ xxd J_ub_law_working.jpg | tail -5
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80 )O.J.Y..(.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45 .....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1 *.-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9 .....Q....

(kali@kali)-[~/Lab3_Data_Carving/working]
$ xxd -l 32 J_ub_law_working.jpg > ../xxd_analysis/jpeg_header.txt
xxd J_ub_law_working.jpg | tail -5 > ../xxd_analysis/jpeg_footer.txt

(kali@kali)-[~/Lab3_Data_Carving/working]
$ cat ../xxd_analysis/jpeg_header.txt
cat ../xxd_analysis/jpeg_footer.txt
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80 )O.J.Y..(.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45 .....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1 *.-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9 .....Q....

(kali@kali)-[~/Lab3_Data_Carving/working]
```

JPEG SOI Start of Image

Your first bytes are: FF D8 FF E1

The critical JPEG SOI marker is: FF D8

So the file begins with a valid JPEG SOI signature.

JPEG EOI End of Image

Your final bytes end: D351 9ebf ffd9. Therefore, FF D9 is present at the end, confirming the JPEG EOI marker.

To create the plain hexadecimal dump,

The xxd utility was used to create a plain hexadecimal representation of the working copy of J_ub_law.jpg. The command used was:

```
xxd -p J_ub_law_working.jpg > ../xxd_analysis/J_ub_law.hex
```

The resulting hexadecimal dump was saved as J_ub_law.hex in the xxd_analysis directory. The beginning of the dump was inspected to confirm that the JPEG data began with the expected JPEG Start of Image (SOI) signature, represented by FF D8.

```
(kali@kali)~[Lab3_Data_Carving/working]
$ xxd -r -p ../xxd_analysis/J_ub_law.hex ../xxd_analysis/J_ub_law_reconstructed.jpg
(kali@kali)~[Lab3_Data_Carving/working]
$ xxd -r -p ../xxd_analysis/J_ub_law.hex ../xxd_analysis/J_ub_law_reconstructed.jpg
(kali@kali)~[Lab3_Data_Carving/working]
$ ls -lh ../xxd_analysis/J_ub_law_reconstructed.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  5 10:19 ../xxd_analysis/J_ub_law_reconstructed.jpg
(kali@kali)~[Lab3_Data_Carving/working]
$ file ../xxd_analysis/J_ub_law_reconstructed.jpg
../xxd_analysis/J_ub_law_reconstructed.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=20
6, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920*1080, components 3
```

Reconstructing the JPEG by Reversing the Plain Hexadecimal Dump

The plain hexadecimal dump was converted back into binary form using the reverse-hexdump functionality of xxd. The command used was:

```
xxd -r -p ../xxd_analysis/J_ub_law.hex ../xxd_analysis/J_ub_law_reconstructed.jpg
```

The reconstructed file was subsequently examined using the file command. The output identified the reconstructed file as a valid JPEG image containing Exif information, confirming that the binary reconstruction was successful.

```
(kali@kali)~[Lab3_Data_Carving/working]
$ xxd -r -p ../xxd_analysis/J_ub_law.hex ../xxd_analysis/J_ub_law_reconstructed.jpg
(kali@kali)~[Lab3_Data_Carving/working]
$ xxd -r -p ../xxd_analysis/J_ub_law.hex ../xxd_analysis/J_ub_law_reconstructed.jpg
(kali@kali)~[Lab3_Data_Carving/working]
$ ls -lh ../xxd_analysis/J_ub_law_reconstructed.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  5 10:19 ../xxd_analysis/J_ub_law_reconstructed.jpg
(kali@kali)~[Lab3_Data_Carving/working]
$ file ../xxd_analysis/J_ub_law_reconstructed.jpg
../xxd_analysis/J_ub_law_reconstructed.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=20
6, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920*1080, components 3
```

The most important part is to compare the hashes

Cryptographic hashes were calculated for both the original working JPEG and the reconstructed JPEG to verify whether the reconstruction produced an identical binary file.

The MD5 hash for both files was:

```
83a360ac7f7e0ca318e5bfe39f95f137
```

The SHA-256 hash for both files was:

```
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468
```

The MD5 and SHA-256 values were identical for the working JPEG and the reconstructed JPEG. This confirms that the reconstructed JPEG is byte-for-byte identical to the working copy and that the hexadecimal dump and reverse-hexdump process successfully reproduced the original binary content.

```
(kali@kali)~/Lab3_Data_Carving/working
$ md5sum J_ub_law_working.jpg ../xxd_analysis/J_ub_law_reconstructed.jpg
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law_working.jpg
83a360ac7f7e0ca318e5bfe39f95f137  ../xxd_analysis/J_ub_law_reconstructed.jpg

(kali@kali)~/Lab3_Data_Carving/working
$ sha256sum J_ub_law_working.jpg ../xxd_analysis/J_ub_law_reconstructed.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_law_working.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  ../xxd_analysis/J_ub_law_reconstructed.jpg

(kali@kali)~/Lab3_Data_Carving/working
$ cd ~/Lab3_Data_Carving/working
$ xxd -p J_ub_law_working.jpg > ../xxd_analysis/J_ub_law.hex
$ xxd -r -p ../xxd_analysis/J_ub_law.hex ../xxd_analysis/J_ub_law_reconstructed.jpg
$ file ../xxd_analysis/J_ub_law_reconstructed.jpg
../xxd_analysis/J_ub_law_reconstructed.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=20]
```

File Signatures and Hex Analysis

Activity	Command Used	Result / Finding
JPEG header analysis	xxd -l 32 J_ub_law_working.jpg	The file begins with FF D8, confirming the JPEG Start of Image (SOI) signature. The output also shows Exif data and identifies the camera as NIKON CORPORATION, model NIKON D4.
JPEG footer analysis	xxd J_ub_law_working.jpg tail -5	The final bytes contain FF D9, confirming the JPEG End of Image (EOI) signature.
Plain hexadecimal dump	xxd -p J_ub_law_working.jpg > ../xxd_analysis/J_ub_law.hex	A plain hexadecimal representation of the JPEG was successfully created and saved as J_ub_law.hex.
JPEG reconstruction	xxd -r -p ../xxd_analysis/J_ub_law.hex > ../xxd_analysis/J_ub_law_reconstructed.jpg	The hexadecimal dump was successfully reversed into a binary JPEG file. The file command confirmed that the reconstructed file is a valid JPEG with Exif information.
MD5 hash comparison	md5sum J_ub_law_working.jpg ../xxd_analysis/J_ub_law_reconstructed.jpg	Both files produced the MD5 hash 83a360ac7f7e0ca318e5bfe39f95f137. The hashes are identical.

Activity	Command Used	Result / Finding
SHA-256 hash comparison	sha256sum J_ub_law_working.jpg ../xxd_analysis/J_ub_law_reconstructed.jpg	Both files produced the SHA-256 hash 238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468. The hashes are identical.

The JPEG was successfully examined at the hexadecimal level and reconstructed from its plain hexadecimal dump. The identical MD5 and SHA-256 hashes demonstrate that the reconstructed JPEG is byte-for-byte identical to the working copy.

Part B: Examine the JPEG content

The strings utility was used to examine the working copy of J_ub_law.jpg for readable character sequences embedded within the binary file. The command `strings -a J_ub_law_working.jpg | tee ../xxd_analysis/jpeg_strings.txt` was executed, and the resulting output was saved to `jpeg_strings.txt` for further examination. This provided a low-level view of printable text contained within the JPEG and allowed potentially relevant metadata-related information to be identified and documented.

```

kali@kali: ~/Lab3_Data_Carving/working
File Actions Edit View Help
sudo: /home/kali/Lab3_Data_Carving/working/J_ub_law_working.jpg: command not found

(kali@kali): ~/Lab3_Data_Carving/working
$ cd ~/Lab3_Data_Carving/working
$ strings -a J_ub_law_working.jpg | tee ../xxd_analysis/jpeg_strings.txt
Exif
NIKON CORPORATION
NIKON D4
Adobe Photoshop 21.1 (Macintosh)
2020:08:20 10:20:05
HOWARD KORN
0221
2013:10:08 18:56:21
2013:10:08 18:56:21
2027756
17.0-35.0 mm f/2.8
Adobe_CM
Adobe
b34r
7Ggw
AQaq"
dEU6te
7Ggw
2VEXK
I:[h
[$wik
S_b[
Subf
kc[ LO
uq\j
Y[h]
["e>
04YC[hakK
X-QI:
12Y:$C
Ydmv/
;/nm
5CuW
ofEO
Nmon=W
JYiS
b"iI
hi0YXk"

```

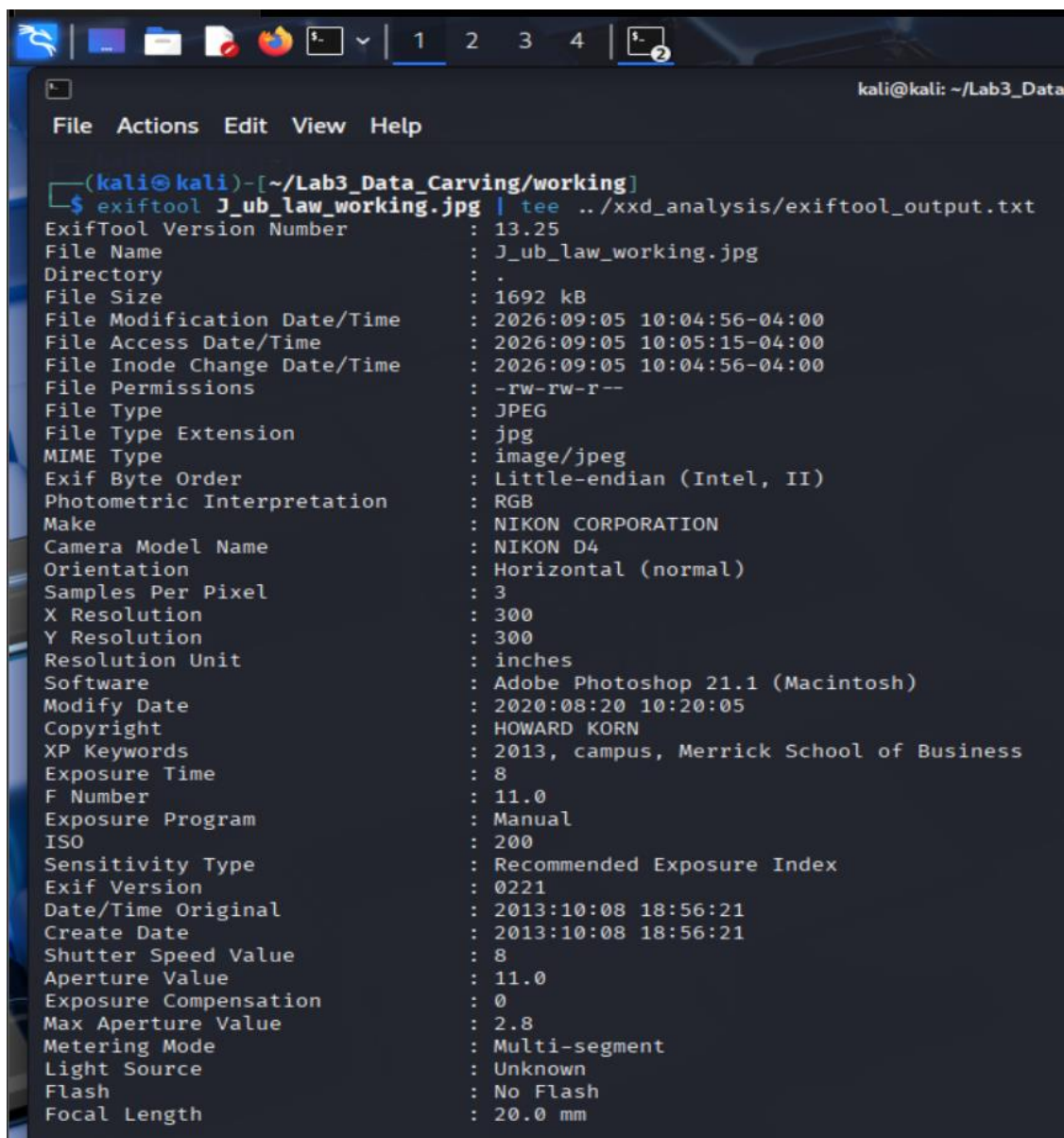
```
kali@kali: ~/Lab3_Data_Carving/working
File Actions Edit View Help
14T12:06:09-05:00"/> <rdf:li stEvt:action="derived" stEvt:parameters="converted from image/jpeg to image/tiff"/> <rdf:li stEvt:action="saved" stEvt:changed=
="/" stEvt:instanceID="xmp.iid:6f2a9c87-e723-419d-9d9b-53ebf8e40a23" stEvt:softwareAgent="Adobe Photoshop Camera Raw 8.3 (Macintosh)" stEvt:when="2014-01-1
5T11:32:49-05:00"/> <rdf:li stEvt:action="derived" stEvt:parameters="converted from image/tiff to image/jpeg"/> <rdf:li stEvt:action="saved" stEvt:changed=
="/" stEvt:instanceID="xmp.iid:E8D48D1A11206811822AFE776815F8C7" stEvt:softwareAgent="Adobe Photoshop CS6 (Macintosh)" stEvt:when="2014-01-15T11:34:04-05:00
"/> <rdf:li stEvt:action="saved" stEvt:instanceID="xmp.iid:5c9dcb11-a9ef-4658-9210-97c98ea2dae3" stEvt:when="2020-05-26T15:08:43-04:00" stEvt:softwareAgent
="Adobe Photoshop 21.1 (Macintosh)" stEvt:changed="/" /> <rdf:li stEvt:action="converted" stEvt:parameters="from image/jpeg to application/vnd.adobe.photosh
op"/> <rdf:li stEvt:action="derived" stEvt:parameters="converted from image/jpeg to application/vnd.adobe.photoshop"/> <rdf:li stEvt:action="saved" stEvt:i
nstanceID="xmp.iid:bab66396-fda3-42d1-8b97-da85ea2e36fb" stEvt:when="2020-05-26T15:08:43-04:00" stEvt:softwareAgent="Adobe Photoshop 21.1 (Macintosh)" stEv
t:changed="/" /> <rdf:li stEvt:action="saved" stEvt:instanceID="xmp.iid:fe897b17-bbe6-4cef-83a4-fdc4accf979" stEvt:when="2020-05-27T16:14:03-04:00" stEvt:s
oftwareAgent="Adobe Photoshop 21.1 (Macintosh)" stEvt:changed="/" /> <rdf:li stEvt:action="converted" stEvt:parameters="from application/vnd.adobe.photoshop
to image/jpeg"/> <rdf:li stEvt:action="derived" stEvt:parameters="converted from application/vnd.adobe.photoshop to image/jpeg"/> <rdf:li stEvt:action="sa
ved" stEvt:instanceID="xmp.iid:ebf6b840-0bea-4644-9e49-50909da8d7db" stEvt:when="2020-05-27T16:14:03-04:00" stEvt:softwareAgent="Adobe Photoshop 21.1 (Maci
ntosh)" stEvt:changed="/" /> <rdf:li stEvt:action="saved" stEvt:instanceID="xmp.iid:66e9f972-24cb-4805-931d-233583583614" stEvt:when="2020-07-02T10:21:18-04:
00" stEvt:softwareAgent="Adobe Photoshop 21.1 (Macintosh)" stEvt:changed="/" /> <rdf:li stEvt:action="converted" stEvt:parameters="from image/jpeg to appli
cation/vnd.adobe.photoshop"/> <rdf:li stEvt:action="derived" stEvt:parameters="converted from image/jpeg to application/vnd.adobe.photoshop"/> <rdf:li stEv
t:action="saved" stEvt:instanceID="xmp.iid:981ec349-e69e-479f-b5d4-600905e0b0e6" stEvt:when="2020-07-02T10:21:18-04:00" stEvt:softwareAgent="Adobe Photosho
p 21.1 (Macintosh)" stEvt:changed="/" /> <rdf:li stEvt:action="saved" stEvt:instanceID="xmp.iid:0bb33b69-4279-4f45-afce-49859bf97ea8" stEvt:when="2020-08-0
7T14:09:42-04:00" stEvt:softwareAgent="Adobe Photoshop 21.1 (Macintosh)" stEvt:changed="/" /> <rdf:li stEvt:action="converted" stEvt:parameters="from applica
tion/vnd.adobe.photoshop to image/jpeg"/> <rdf:li stEvt:action="derived" stEvt:parameters="converted from application/vnd.adobe.photoshop to image/jpeg"/>
<rdf:li stEvt:action="saved" stEvt:instanceID="xmp.iid:3f0c89f0-f710-45fe-a8f8-01d5e1acc22f" stEvt:when="2020-08-17T14:09:42-04:00" stEvt:softwareAgent="
Adobe Photoshop 21.1 (Macintosh)" stEvt:changed="/" /> <rdf:li stEvt:action="saved" stEvt:instanceID="xmp.iid:9e7f6b27-04e9-43e7-ae79-c3f2fb9c2ac9" stEvt:wh
en="2020-08-20T10:20:05-04:00" stEvt:softwareAgent="Adobe Photoshop 21.1 (Macintosh)" stEvt:changed="/" /> <rdf:Seq> </xmpMM:History> <crs:ToneCurvePV2012>
rdf:Seq> <rdf:li>0, 0</rdf:li> <rdf:li>255, 255</rdf:li> </rdf:Seq> </crs:ToneCurvePV2012> <crs:ToneCurvePV2012Blue> <rdf:Seq> <rdf:li>0, 0</rdf:li> <rdf
:li>255, 255</rdf:li> </rdf:Seq> </crs:ToneCurvePV2012Blue> <crs:ToneCurvePV2012Green> <rdf:Seq> <rdf:li>0, 0</rdf:li> <rdf:li>255, 255</rdf:li> </rdf:Seq>
</crs:ToneCurvePV2012Green> <crs:ToneCurvePV2012Red> <rdf:Seq> <rdf:li>0, 0</rdf:li> <rdf:li>255, 255</rdf:li> </rdf:Seq> </crs:ToneCurvePV2012Red> <lr:hier
archicalSubject> <rdf:Bag> <rdf:li>Schools|Merrick School of Business</rdf:li> </rdf:Bag> </lr:hierarchicalSubject> </rdf:Description> </rdf:RDF> </x:xmpme
ta>

<?xpacket end="w"?>

@ICC_PROFILE
0ADBE
mnrRGB XYZ
```

Targeted Metadata String Search

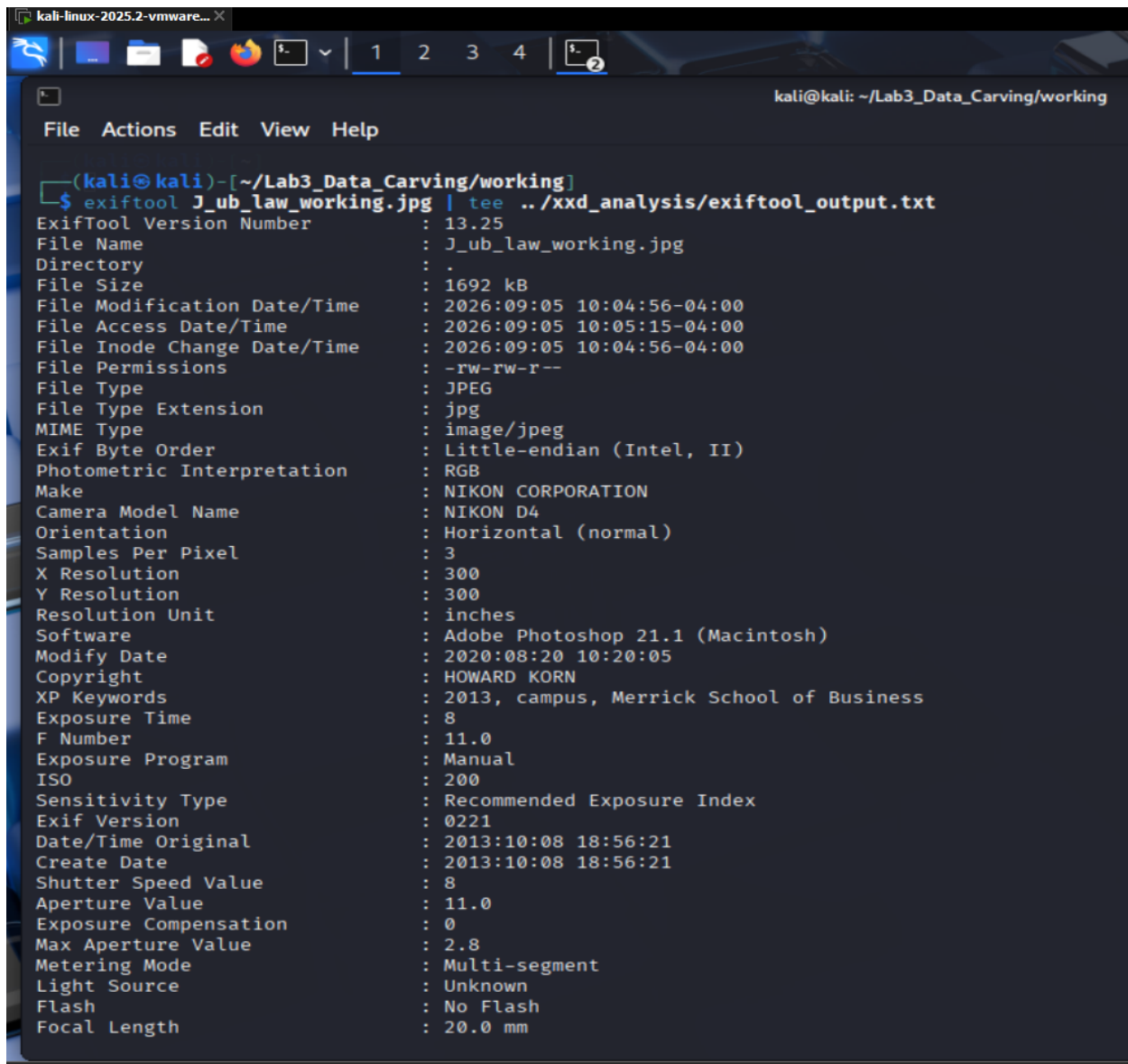
A targeted string search was subsequently performed to identify readable information relating to the camera, software, date, author, copyright, GPS and other potentially relevant metadata. The command strings `-a J_ub_law_working.jpg | grep -Ei 'NIKON|Nikon|camera|software|date|author|copyright|GPS|Adobe|Photoshop|name'` | tee ../xxd_analysis/jpeg_metadata_strings.txt was used to filter the extracted strings for the specified terms. The resulting output was saved as jpeg_metadata_strings.txt. Only information actually returned by the examination was considered as evidence, while metadata categories for which no corresponding output was observed were recorded as “Not found” rather than inferred.



```
(kali@kali)-[~/Lab3_Data_Carving/working]
$ exiftool J_ub_law_working.jpg | tee ../xtd_analysis/exiftool_output.txt
ExifTool Version Number      : 13.25
File Name                    : J_ub_law_working.jpg
Directory                    : .
File Size                    : 1692 kB
File Modification Date/Time   : 2026:09:05 10:04:56-04:00
File Access Date/Time        : 2026:09:05 10:05:15-04:00
File Inode Change Date/Time   : 2026:09:05 10:04:56-04:00
File Permissions              : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Little-endian (Intel, II)
Photometric Interpretation    : RGB
Make                         : NIKON CORPORATION
Camera Model Name             : NIKON D4
Orientation                   : Horizontal (normal)
Samples Per Pixel             : 3
X Resolution                  : 300
Y Resolution                  : 300
Resolution Unit               : inches
Software                      : Adobe Photoshop 21.1 (Macintosh)
Modify Date                   : 2020:08:20 10:20:05
Copyright                    : HOWARD KORN
XP Keywords                   : 2013, campus, Merrick School of Business
Exposure Time                 : 8
F Number                      : 11.0
Exposure Program              : Manual
ISO                           : 200
Sensitivity Type              : Recommended Exposure Index
Exif Version                  : 0221
Date/Time Original            : 2013:10:08 18:56:21
Create Date                   : 2013:10:08 18:56:21
Shutter Speed Value           : 8
Aperture Value                : 11.0
Exposure Compensation         : 0
Max Aperture Value            : 2.8
Metering Mode                 : Multi-segment
Light Source                  : Unknown
Flash                        : No Flash
Focal Length                  : 20.0 mm
```

```
kali-linux-2025.2-vmware... X
File Actions Edit View Help
File Source : Digital Camera
Scene Type : Directly photographed
CFA Pattern : [Red,Green][Green,Blue]
Custom Rendered : Normal
Exposure Mode : Manual
White Balance : Auto
Digital Zoom Ratio : 1
Focal Length In 35mm Format : 20 mm
Scene Capture Type : Standard
Gain Control : None
Contrast : Normal
Saturation : Normal
Sharpness : Normal
Subject Distance Range : Unknown
Serial Number : 2027750
Lens Info : 17-35mm f/2.8
Lens Model : 17.0-35.0 mm f/2.8
Compression : JPEG (old-style)
Thumbnail Offset : 1198
Thumbnail Length : 4934
Current IPTC Digest : 70e1212f6f80a22b8acca88b2b4633ef
Envelope Record Version : 4
Coded Character Set : UTF8
Application Record Version : 2
Time Created : 00:00:00
Keywords : 2013, campus, Merrick School of Business
Copyright Notice : HOWARD KORN
IPTC Digest : 70e1212f6f80a22b8acca88b2b4633ef
Displayed Units X : inches
Displayed Units Y : inches
Print Style : Centered
Print Position : 0 0
Print Scale : 1
Global Angle : 30
Global Altitude : 30
URL List :
Slices Group Name : Law_1_zoom
Num Slices : 1
Pixel Aspect Ratio : 1
Photoshop Thumbnail : (Binary data 4934 bytes, use -b option to extract)
Has Real Merged Data : Yes
Writer Name : Adobe Photoshop
Reader Name : Adobe Photoshop 2020
```

The JPEG content was examined using strings, a targeted metadata-related string search and ExifTool. The examination confirmed that the image contains embedded Exif metadata and provided additional information that could be used to characterise the image and its source. The camera information previously identified from the file examination, including NIKON CORPORATION as the manufacturer and NIKON D4 as the camera model, was consistent with the metadata examination. All additional findings were recorded directly from the examination results, while information that could not be established from the available evidence was reported as “Not found.”



```
(kali@kali)-[~/Lab3_Data_Carving/working]
$ exiftool J_ub_law_working.jpg | tee ../xxd_analysis/exiftool_output.txt
ExifTool Version Number      : 13.25
File Name                    : J_ub_law_working.jpg
Directory                    : .
File Size                    : 1692 kB
File Modification Date/Time   : 2026:09:05 10:04:56-04:00
File Access Date/Time        : 2026:09:05 10:05:15-04:00
File Inode Change Date/Time   : 2026:09:05 10:04:56-04:00
File Permissions              : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Little-endian (Intel, II)
Photometric Interpretation    : RGB
Make                         : NIKON CORPORATION
Camera Model Name            : NIKON D4
Orientation                   : Horizontal (normal)
Samples Per Pixel             : 3
X Resolution                  : 300
Y Resolution                  : 300
Resolution Unit               : inches
Software                     : Adobe Photoshop 21.1 (Macintosh)
Modify Date                   : 2020:08:20 10:20:05
Copyright                    : HOWARD KORN
XP Keywords                   : 2013, campus, Merrick School of Business
Exposure Time                 : 8
F Number                     : 11.0
Exposure Program              : Manual
ISO                           : 200
Sensitivity Type              : Recommended Exposure Index
Exif Version                  : 0221
Date/Time Original            : 2013:10:08 18:56:21
Create Date                   : 2013:10:08 18:56:21
Shutter Speed Value           : 8
Aperture Value                : 11.0
Exposure Compensation         : 0
Max Aperture Value            : 2.8
Metering Mode                 : Multi-segment
Light Source                  : Unknown
Flash                        : No Flash
Focal Length                  : 20.0 mm
```

Part C: The Sleuth Kit

This is the section where we'll analyze Ch01InChap01.dd, the forensic dd image to determine the actual partition/sector/block values from your own output.

```
(kali@kali)-[~/Lab3_Data_Carving/working]
$ which img_stat
which mmls
which fsstat
which fls
which icat
which blkcat
/usr/bin/img_stat
/usr/bin/mmls
/usr/bin/fsstat
/usr/bin/fls
/usr/bin/icat
/usr/bin/blkcat
```

The `img_stat` analysis confirmed that `Ch01InChap01.dd` is a raw forensic image with a total size of 1,474,560 bytes and a sector size of 512 bytes. This establishes the basic characteristics of the evidence image before filesystem examination.

```
(kali@kali)-[~/Lab3_Data_Carving/working]
$ cd ~/Lab3_Data_Carving/original_evidence
file Ch01InChap01.dd
ls -lh Ch01InChap01.dd
Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID ", 'k0nIHC" cached by Windows 9M, root entries 224, sectors 2880 (volumes ≤ 32 MB), sectors /FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT
-rw-rw-r-- 1 kali kali 1.5M Sep  5 09:58 Ch01InChap01.dd

(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ img_stat Ch01InChap01.dd | tee ~/Lab3_Data_Carving/tsk_analysis/img_stat.txt
IMAGE FILE INFORMATION
-----
Image Type: raw
Size in bytes: 1474560
Sector size: 512
```

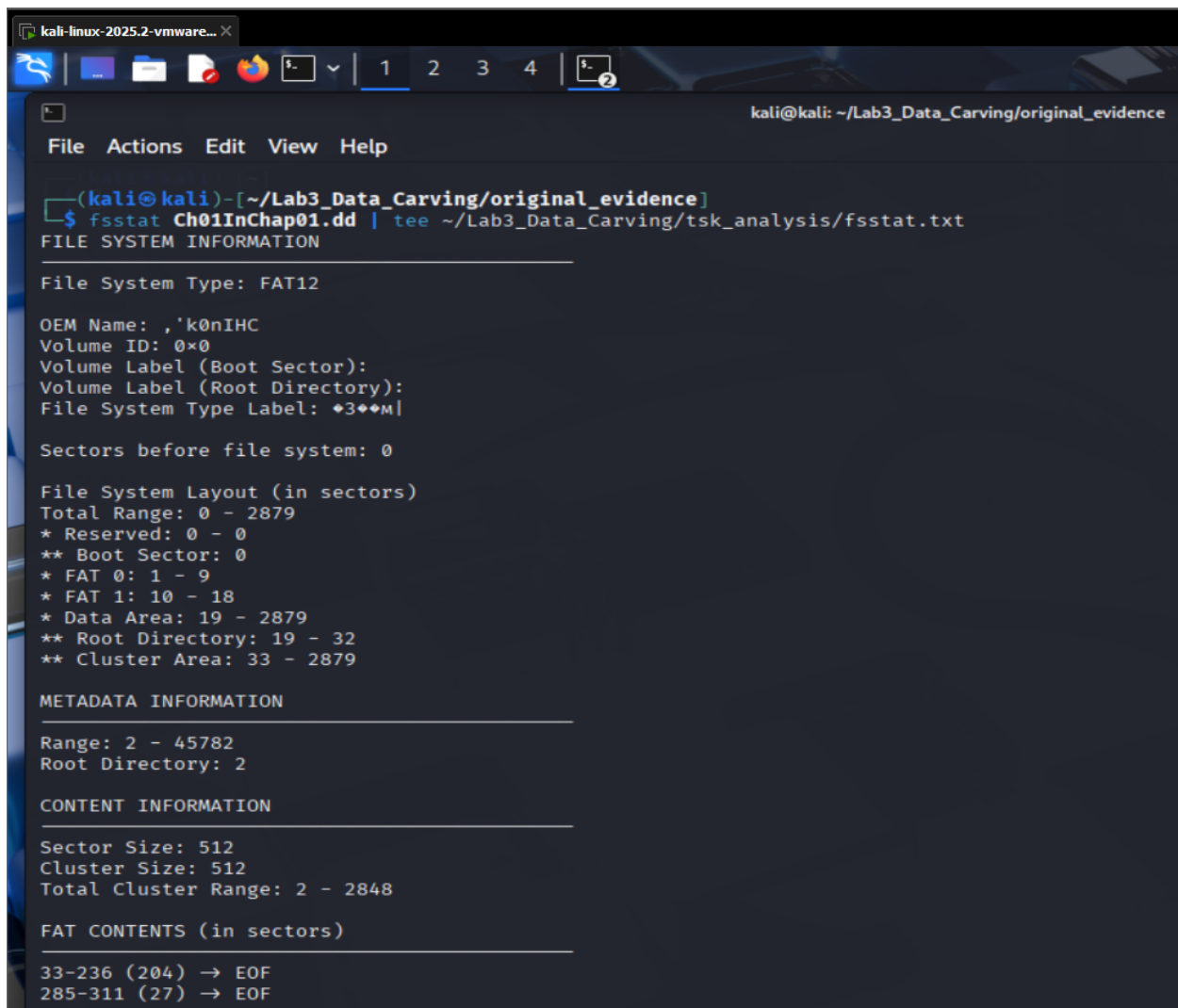
The `mmls` command produced no partition-table output, indicating that the image does not contain a conventional partition table. This is consistent with the subsequent `fsstat` results, which show that the FAT12 filesystem begins at sector 0.

```
(kali@kali)-[~/Lab3_Data_Carving/working]
$ cd ~/Lab3_Data_Carving/original_evidence
file Ch01InChap01.dd
ls -lh Ch01InChap01.dd
Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID ", 'k0nIHC" cached by Windows 9M, root entries 224, sectors 2880 (volumes ≤ 32 MB), sectors /FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT
-rw-rw-r-- 1 kali kali 1.5M Sep  5 09:58 Ch01InChap01.dd

(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ img_stat Ch01InChap01.dd | tee ~/Lab3_Data_Carving/tsk_analysis/img_stat.txt
IMAGE FILE INFORMATION
-----
Image Type: raw
Size in bytes: 1474560
Sector size: 512

(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ mmls Ch01InChap01.dd | tee ~/Lab3_Data_Carving/tsk_analysis/mmls.txt
```

The recursive `fls` examination identified six files within the FAT12 filesystem. Four entries - `Billing Letter.doc`, `confirmation.txt`, `letter1.txt`, and `Regrets.doc`—are marked with an asterisk, indicating deleted directory entries. The remaining files, `Client Info.mdb` and `Income.xls`, are listed as active files. The identified deleted files provide potential targets for forensic recovery using The Sleuth Kit.



```
kali@kali: ~/Lab3_Data_Carving/original_evidence
File Actions Edit View Help

(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ fsstat Ch01InChap01.dd | tee ~/Lab3_Data_Carving/tsk_analysis/fsstat.txt
FILE SYSTEM INFORMATION
-----
File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: ♦3♦M|

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)
-----
33-236 (204) → EOF
285-311 (27) → EOF
```

What image result confirmed the following:

- Raw image: 1,474,560 bytes
- Sector size: 512 bytes
- Filesystem: FAT12
- No partition offset: Sectors before file system: 0
- Root directory: sectors 19–32
- Cluster area: sectors 33–2879

Most importantly, fls found six files:

```
(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
$ fls -r -p Ch01InChap01.dd | tee ~/Lab3_Data_Carving/tsk_analysis/fls_recursive.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

Entry	Status	File
5	Normal	Client Info.mdb
8	*	Billing Letter.doc
11	*	confirmation.txt
13	Normal	Income.xls
15	*	letter1.txt
17	*	Regrets.doc

The * entries are particularly interesting because they indicate deleted files. This is exactly where the forensic recovery/carving portion becomes useful.

Part C: Recover the deleted files

The Sleuth Kit was used to examine the forensic image Ch01InChap01.dd and identify files present within the FAT12 filesystem. The fls command was used to recursively list the filesystem contents. The results identified six files, including four entries marked with an asterisk, indicating that the files had been deleted. The deleted files identified were Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. The screenshot below shows the output of the fls command and the identified deleted files.

```
(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
$ fls -r -p Ch01InChap01.dd | tee ~/Lab3_Data_Carving/tsk_analysis/fls_recursive.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

The deleted files were subsequently recovered from the forensic image using the Sleuth Kit `icat` utility. Each deleted file was recovered using its corresponding metadata address identified from the `fls` output and saved into a separate recovered directory. This approach ensured that the original forensic image remained unchanged while allowing further examination of the recovered evidence. The successful execution of the recovery commands is shown in the screenshot below.

```
(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
$ mkdir -p ~/Lab3_Data_Carving/recovered

(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
$ icat -r Ch01InChap01.dd 8 > ~/Lab3_Data_Carving/recovered/Billing_Letter.doc

(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
$ icat -r Ch01InChap01.dd 11 > ~/Lab3_Data_Carving/recovered/confirmation.txt

(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
$ icat -r Ch01InChap01.dd 15 > ~/Lab3_Data_Carving/recovered/letter1.txt

(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
$ icat -r Ch01InChap01.dd 17 > ~/Lab3_Data_Carving/recovered/Regrets.doc
```

Following recovery, the `ls -lh` command was used to verify that the recovered files had been successfully created and to document their file sizes. Four files were recovered: `Billing_Letter.doc` (24 KB), `confirmation.txt` (227 bytes), `letter1.txt` (121 bytes), and `Regrets.doc` (23 KB). The screenshot below demonstrates the presence and sizes of the recovered files.

```
(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
$ ls -lh ~/Lab3_Data_Carving/recovered/
total 56K
-rw-rw-r-- 1 kali kali 24K Sep  5 10:59 Billing_Letter.doc
-rw-rw-r-- 1 kali kali 227 Sep  5 10:59 confirmation.txt
-rw-rw-r-- 1 kali kali 121 Sep  5 10:59 letter1.txt
-rw-rw-r-- 1 kali kali 23K Sep  5 11:00 Regrets.doc
```

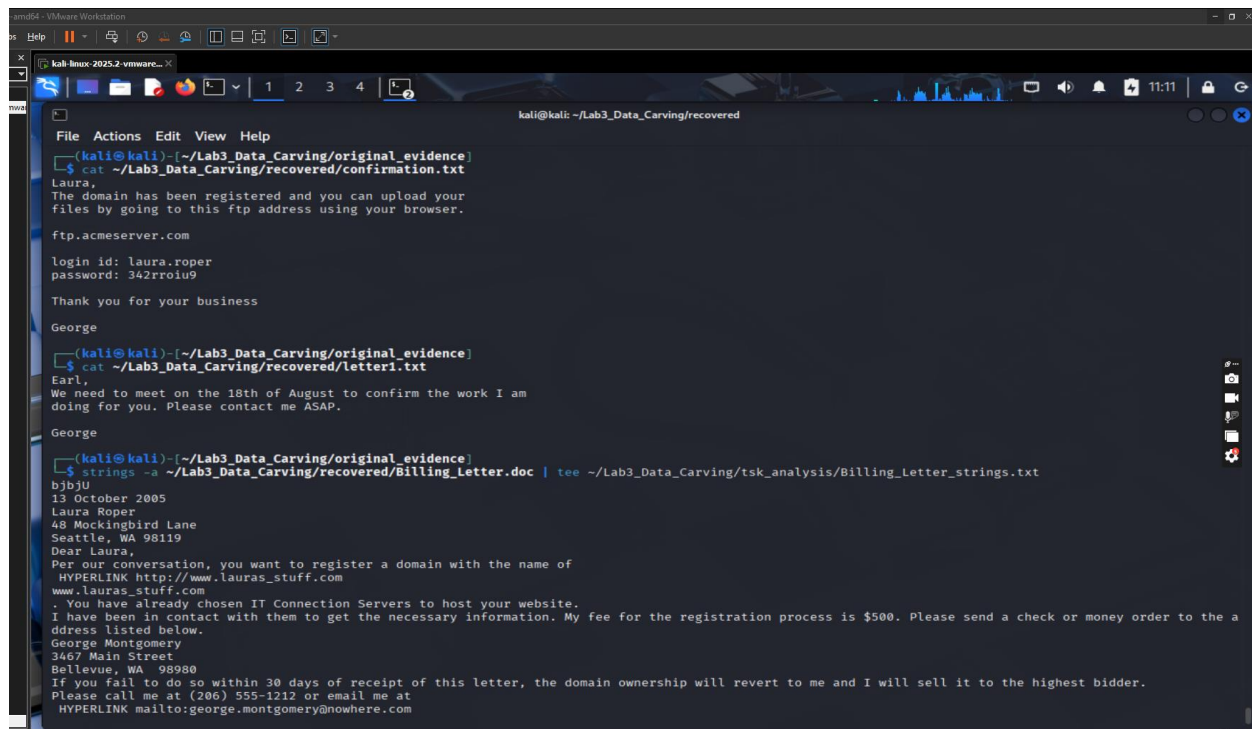
The `file` command was then used to determine the file types and examine available document metadata. The two recovered `.doc` files were identified as Microsoft Composite Document File V2 documents. `Billing_Letter.doc` contained metadata identifying the recorded author as Amelia Phillips, with a document title of 13 October 2003, while `Regrets.doc` contained the recorded author as Amelia Phillips and a document title of 02 November 2003. The metadata also provided information such as the creating application, revision number, creation time, last saved time, page count, and word count. The screenshot below shows the file identification and metadata recovered from the two Microsoft Word documents.

```
(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
$ file ~/Lab3_Data_Carving/recovered/*
/home/kali/Lab3_Data_Carving/recovered/Billing_Letter.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252,
Title: 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Mi
crosoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri Dec  9 14:50:00 2005, Number of Pages: 1
, Number of Words: 118, Number of Characters: 678, Security: 0
```

The recovery results demonstrate that deleted files can remain recoverable from a filesystem when their underlying data has not been overwritten. The examination also demonstrates the value of combining filesystem analysis with file identification, as recovering a deleted file can provide both the original file content and potentially useful embedded metadata for forensic examination.

Next is to examine the recovered text as the actual contents will now be preserved as evidence rather than opening them graphically

The recovered files were examined using the file utility to validate their file types and identify available embedded metadata. The results confirmed that Billing_Letter.doc and Regrets.doc were Microsoft Word Compound Document files, while confirmation.txt and letter1.txt were identified as ASCII text files. The Word documents also contained embedded metadata, including Amelia Phillips as the recorded author and last saved-by user, Microsoft Word 9.0 as the creating application, and revision information. The metadata was recorded as documentary evidence and was not treated independently as proof of the physical identity of the person who created the documents.



```
kali@kali: ~/Lab3_Data_Carving/recovered
File Actions Edit View Help
(kali@kali)~[~/Lab3_Data_Carving/original_evidence]
$ cat ~/Lab3_Data_Carving/recovered/confirmation.txt
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

Login id: laura.roper
password: 342rroi9

Thank you for your business

George

(kali@kali)~[~/Lab3_Data_Carving/original_evidence]
$ cat ~/Lab3_Data_Carving/recovered/letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(kali@kali)~[~/Lab3_Data_Carving/original_evidence]
$ strings -a ~/Lab3_Data_Carving/recovered/Billing_Letter.doc | tee ~/Lab3_Data_Carving/tsk_analysis/Billing_Letter_strings.txt
bjbjU
13 October 2005
Laura Roper
48 Mockingbird Lane
Seattle, WA 98119
Dear Laura,
Per our conversation, you want to register a domain with the name of
HYPERLINK http://www.lauras_stuff.com
www.lauras_stuff.com
. You have already chosen IT Connection Servers to host your website.
I have been in contact with them to get the necessary information. My fee for the registration process is $500. Please send a check or money order to the a
dress listed below.
George Montgomery
3467 Main Street
Bellevue, WA 98980
If you fail to do so within 30 days of receipt of this letter, the domain ownership will revert to me and I will sell it to the highest bidder.
Please call me at (206) 555-1212 or email me at
HYPERLINK mailto:george.montgomery@nowhere.com
```

```
id64 - VMware Workstation
kali-linux-2025.2-vmware...
kali@kali: ~/Lab3_Data_Carving/recovered
File Actions Edit View Help
(kali@kali)~/Lab3_Data_Carving/original_evidence
$ cat ~/Lab3_Data_Carving/recovered/letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.
George
(kali@kali)~/Lab3_Data_Carving/original_evidence
$ strings -a ~/Lab3_Data_Carving/recovered/Billing_Letter.doc | tee ~/Lab3_Data_Carving/tsk_analysis/Billing_Letter_strings.txt
bjbjU
13 October 2005
Laura Roper
48 Mockingbird Lane
Seattle, WA 98119
Dear Laura,
Per our conversation, you want to register a domain with the name of
HYPERLINK http://www.lauras_stuff.com
www.lauras_stuff.com
. You have already chosen IT Connection Servers to host your website.
I have been in contact with them to get the necessary information. My fee for the registration process is $500. Please send a check or money order to the
address listed below.
George Montgomery
3467 Main Street
Bellevue, WA 98980
If you fail to do so within 30 days of receipt of this letter, the domain ownership will revert to me and I will sell it to the highest bidder.
Please call me at (206) 555-1212 or email me at
HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
Regards,
George Montgomery
13 October 2003
Amelia Phillips
Normal.dot
Amelia Phillips
Microsoft Word 9.0
Starships CMS
13 October 2003
Title
PID_HLINKS
Microsoft Word Document
MSWordDoc
Word.Document.8
```

The recovered confirmation.txt file contained correspondence addressed to Laura concerning a registered domain and instructions for uploading files to an FTP server. The message included an FTP server address, login information and authentication credentials, and was signed by George. The recovered content demonstrates that the deleted file contained information relating to an FTP account and file-transfer activity. Because the file contains authentication credentials, the credentials should be treated as sensitive evidence and should only be reproduced in the report where specifically required by the assessment.

```
(kali@kali)~/Lab3_Data_Carving/original_evidence
$ cat ~/Lab3_Data_Carving/recovered/confirmation.txt
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroi9

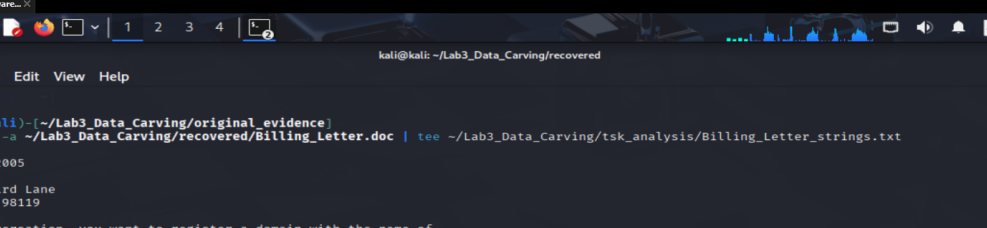
Thank you for your business
George
(kali@kali)~/Lab3_Data_Carving/original_evidence
$ cat ~/Lab3_Data_Carving/recovered/letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.
George
(kali@kali)~/Lab3_Data_Carving/original_evidence
$ strings -a ~/Lab3_Data_Carving/recovered/Billing_Letter.doc | tee ~/Lab3_Data_Carving/tsk_analysis/Billing_Letter_strings.txt
bjbjU
13 October 2005
Laura Roper
48 Mockingbird Lane
Seattle, WA 98119
Dear Laura,
Per our conversation, you want to register a domain with the name of
HYPERLINK http://www.lauras_stuff.com
www.lauras_stuff.com
. You have already chosen IT Connection Servers to host your website.
I have been in contact with them to get the necessary information. My fee for the registration process is $500. Please send a check or money order to the a
dress listed below.
```

```

(kali㉿kali)-[~/Lab3_Data_Carving/original_evidence]
└─$ cat ~/Lab3_Data_Carving/recovered/letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

```



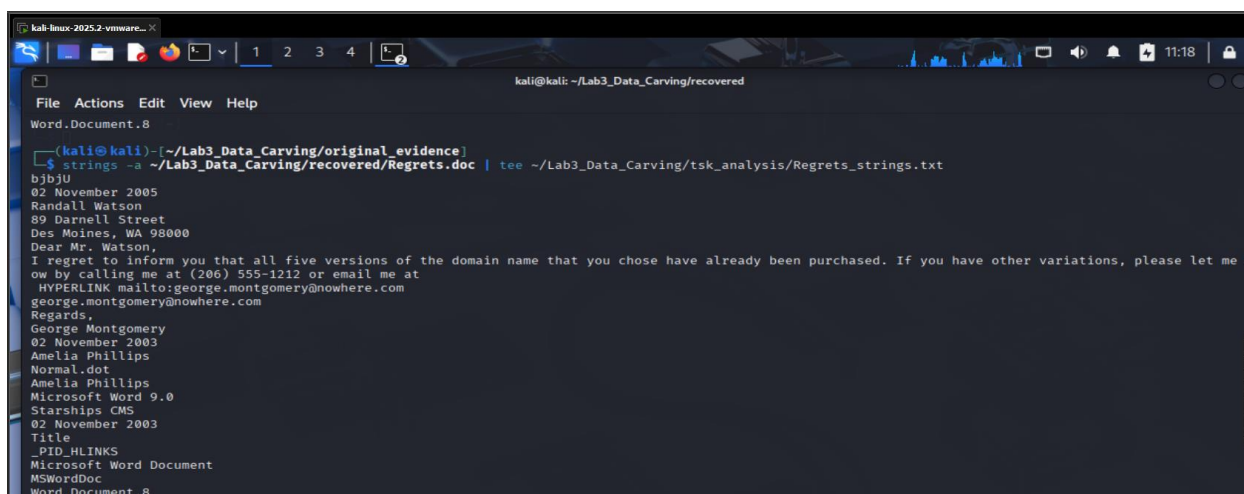
```

kali@kali: ~/Lab3_Data_Carving/recovered
File Actions Edit View Help
George
(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ strings -a ~/Lab3_Data_Carving/recovered/Billing_Letter.doc | tee ~/Lab3_Data_Carving/tsk_analysis/Billing_Letter_strings.txt
b1b1u
13 October 2005
Laura Roper
48 Mockingbird Lane
Seattle, WA 98119
Dear Laura,
Per our conversation, you want to register a domain with the name of
HYPERLINK http://www.lauras_stuff.com
www.lauras_stuff.com
. You have already chosen IT Connection Servers to host your website.
I have been in contact with them to get the necessary information. My fee for the registration process is $500. Please send a check or money order to the a
dress listed below.
George Montgomery
3467 Main Street
Bellevue, WA 98190
If you fail to do so within 30 days of receipt of this letter, the domain ownership will revert to me and I will sell it to the highest bidder.
Please call me at (206) 555-1212 or email me at
HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
Regards,
George Montgomery
13 October 2003
Amelia Phillips
Normal.dot
Amelia Phillips
Microsoft Word 9.0
Starships CMS
13 October 2003
Title
_PID_HLINKS
Microsoft Word Document
MSWordDoc
Word.Document.8

```

Examination of Regrets.doc

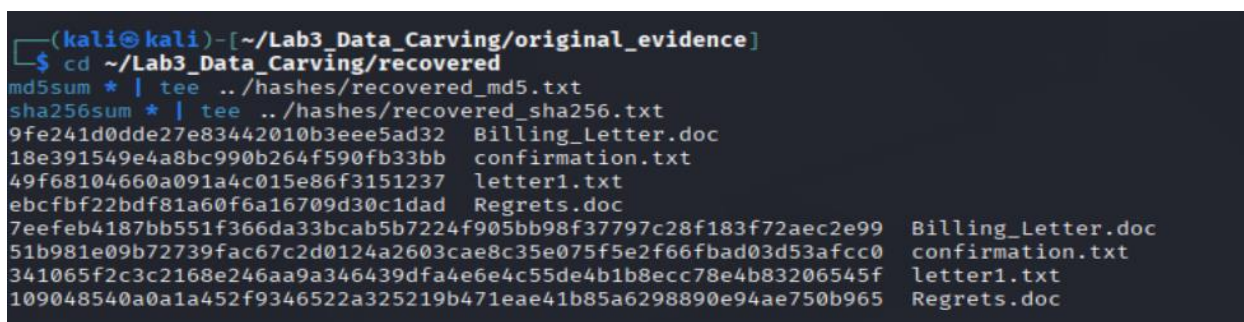
String analysis of the recovered Regrets.doc revealed correspondence addressed to Randall Watson concerning previously purchased domain-name variations. The document states that five versions of the requested domain name had already been purchased and asks the recipient to provide alternative variations. The document also contains contact information associated with George Montgomery. Embedded Microsoft Word metadata identifies Amelia Phillips as the recorded author and last saved-by user and identifies Microsoft Word 9.0 as the creating application. A date discrepancy was also observed, with the document content showing 02 November 2005 while the embedded metadata records 02 November 2003. The discrepancy was documented as an evidentiary observation without assuming its cause.



```
kali@kali: ~/Lab3_Data_Carving/recovered
File Actions Edit View Help
Word.Document.8
(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ strings -a ~/Lab3_Data_Carving/recovered/Regrets.doc | tee ~/Lab3_Data_Carving/tsk_analysis/Regrets_strings.txt
bjbju
02 November 2005
Randall Watson
89 Darnell Street
Des Moines, WA 98000
Dear Mr. Watson,
I regret to inform you that all five versions of the domain name that you chose have already been purchased. If you have other variations, please let me know by calling me at (206) 555-1212 or email me at
HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
Regards,
George Montgomery
02 November 2003
Amelia Phillips
Normal.dot
Amelia Phillips
Microsoft Word 9.0
Starships CMS
02 November 2003
Title
_PID_HLINKS
Microsoft Word Document
MSWordDoc
Word.Document.8
```

Cryptographic Hashing of Recovered Evidence

MD5 and SHA-256 cryptographic hashes were generated for all four recovered files after recovery. The hashes provide unique identifiers for the recovered artefacts and establish baseline integrity values for subsequent examination and reporting. The MD5 and SHA-256 values were recorded in separate hash files within the case directory. These values can be used to verify that the recovered files remain unchanged during the remainder of the forensic examination.



```
(kali@kali)-[~/Lab3_Data_Carving/original_evidence]
$ cd ~/Lab3_Data_Carving/recovered
md5sum * | tee ../hashes/recovered_md5.txt
sha256sum * | tee ../hashes/recovered_sha256.txt
9fe241d0dde27e83442010b3eee5ad32 Billing_Letter.doc
18e391549e4a8bc990b264f590fb33bb confirmation.txt
49f68104660a091a4c015e86f3151237 letter1.txt
ebcfbf22bdf81a60f6a16709d30c1dad Regrets.doc
7eefeb4187bb551f366da33bcab5b7224f905bb98f37797c28f183f72aec2e99 Billing_Letter.doc
51b981e09b72739fac67c2d0124a2603cae8c35e075f5e2f66fbad03d53afcc0 confirmation.txt
341065f2c3c2168e246aa9a346439dfa4e6e4c55de4b1b8ecc78e4b83206545f letter1.txt
109048540a0a1a452f9346522a325219b471eae41b85a6298890e94ae750b965 Regrets.doc
```

Recovered file

File	MD5
Billing_Letter.doc	9fe241d0dde27e83442010b3eee5ad32
confirmation.txt	18e391549e4a8bc990b264f590fb33bb
letter1.txt	49f68104660a091a4c015e86f3151237
Regrets.doc	ebcfbf22bdf81a60f6a16709d30c1dad

File	SHA-256
Billing_Letter.doc	7eefeb4187bb551f366da33bcab5b7224f905bb98f37797c28f183f72aec2e99
confirmation.txt	51b981e09b72739fac67c2d0124a2603cae8c35e075f5e2f66fbad03d53afcc0
letter1.txt	341065f2c3c2168e246aa9a346439dfa4e6e4c55de4b1b8ecc78e4b83206545f
Regrets.doc	109048540a0a1a452f9346522a325219b471eae41b85a6298890e94ae750b965

The recovered files directory was examined to confirm the output of the file recovery process. The `ls -lh recovered/` command shows that four files were successfully recovered from the forensic image: `Billing_Letter.doc`, `confirmation.txt`, `letter1.txt`, and `Regrets.doc`. The recovered files have different sizes, with the two Microsoft Word documents being approximately 23–24 KB and the text files being significantly smaller. This confirms that the recovery process produced distinct files that could be subjected to further forensic examination.

```
(kali㉿kali)-[~/Lab3_Data_Carving/recovered]
$ cd ~/Lab3_Data_Carving

(kali㉿kali)-[~/Lab3_Data_Carving]
$ ls -lh recovered/ | tee tsk_analysis/recovered_files.txt
total 56K
-rw-rw-r-- 1 kali kali 24K Sep  5 10:59 Billing_Letter.doc
-rw-rw-r-- 1 kali kali 227 Sep  5 10:59 confirmation.txt
-rw-rw-r-- 1 kali kali 121 Sep  5 10:59 letter1.txt
-rw-rw-r-- 1 kali kali 23K Sep  5 11:00 Regrets.doc
```

The file command was used to determine the type and basic metadata of each recovered file. The results identified Billing_Letter.doc and Regrets.doc as Microsoft Compound Document/Word documents, while confirmation.txt and letter1.txt were identified as ASCII text files. The output also revealed metadata associated with the Word documents, including author information, document titles, creation and modification timestamps, and document statistics. This step helped establish that the recovered data consisted of valid, recognizable file types rather than unidentified or corrupted data.

Important forensic observation:

The Word documents contain date discrepancies between the visible document content and the metadata. Keep that observation for your final findings rather than trying to resolve it or assume which date is correct.

```
(kali@kali)-[~/Lab3_Data_Carving]
$ file recovered/* | tee tsk_analysis/recovered_file_types.txt
recovered/Billing_Letter.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0
recovered/confirmation.txt: ASCII text, with CRLF line terminators
recovered/letter1.txt: ASCII text, with CRLF line terminators
recovered/Regrets.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 02 November 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 02:00, Create Time/Date: Sat Nov 23 03:12:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 58, Number of Characters: 336, Security: 0
```

Deleted File Identification with Sleuth Kit

The Sleuth Kit fls command was used recursively to examine the FAT12 filesystem and identify files and directories, including deleted entries. The output identified five user files, of which four were marked with an asterisk (*), indicating that they had been deleted. These were Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. The filesystem also contained non-deleted files including Client Info.mdb and Income.xls. The \$MBR, \$FAT1, \$FAT2, and \$OrphanFiles entries represent filesystem structures identified by Sleuth Kit. The four deleted files were subsequently recovered using their metadata/file identifiers with icat.

```
(kali@kali)-[~/Lab3_Data_Carving]
$ fls -r -p original_evidence/Ch01InChap01.dd | tee tsk_analysis/fls_deleted_files.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
v/v 45782: $OrphanFiles
```

This is a particularly important because it provides the forensic basis for saying the four files were deleted files rather than simply files found somewhere on the image.

Binwalk Analysis of the JPEG

Binwalk was used to examine the JPEG working copy for embedded file signatures and other recognizable structures. The analysis identified the file as JPEG image data containing EXIF information at offset 0x0. A TIFF structure associated with the EXIF

metadata was identified at offset 0xC, and a copyright string belonging to Adobe Systems Incorporated was identified at decimal offset 26844 (0x68DC). No additional embedded archive or file signatures were reported by Binwalk. These results are consistent with the earlier hexadecimal and metadata analysis of the JPEG.

```
(kali@kali)~[~/Lab3_Data_Carving]
$ binwalk working/J_ub_low_working.jpg | tee xxd_analysis/binwalk_jpeg.txt
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data, little-endian offset of first image directory: 8
26844	0x68DC	Copyright string: "Copyright 1999 Adobe Systems Incorporated"

Binwalk Analysis of the Forensic Image

Binwalk was also executed against Ch01InChap01.dd to search for recognizable embedded file signatures. The analysis returned no detected signatures. This does not indicate that the forensic image contained no files; rather, it indicates that Binwalk did not identify additional embedded structures using its signature-based scanning. The presence of recoverable files was instead established through filesystem analysis with Sleuth Kit, which identified the FAT12 filesystem and the deleted file entries.

Binwalk Analysis of the Forensic Image

Binwalk was also executed against Ch01InChap01.dd to search for recognizable embedded file signatures. The analysis returned no detected signatures. This does not indicate that the forensic image contained no files; rather, it indicates that Binwalk did not identify additional embedded structures using its signature-based scanning. The presence of recoverable files was instead established through filesystem analysis with Sleuth Kit, which identified the FAT12 filesystem and the deleted file entries.

Confirming Scalpel Installation

The which scalpel command was used to verify that the Scalpel file-carving utility was installed and available on the forensic workstation. The command returned /usr/bin/scalpel, confirming that Scalpel was successfully installed and could be executed from the command line. This verification was performed before configuring the tool and beginning the independent file-carving stage of the investigation.

Reviewing the Scalpel Configuration

The default Scalpel configuration file was examined using grep to determine which file types were currently enabled for carving. The results showed that the JPEG and Microsoft Word (doc) carving signatures were present in the configuration file but were commented out, as indicated by the # characters. The configuration included signatures for JPEG files beginning with JPEG/EXIF or JFIF headers and Microsoft Word documents using the Compound File Binary Format signature. This review was necessary to identify the relevant signatures that needed to be enabled for the forensic image analysis.

```
(kali@kali)-[~/Lab3_Data_Carving]
$
(kali@kali)-[~/Lab3_Data_Carving]
$ grep -E '^(\doc|txt|jpg|jpeg|pdf|png|gif)' /etc/scalpel/scalpel.conf

(kali@kali)-[~/Lab3_Data_Carving]
$ grep -n -A5 -B2 -E 'jpg|doc|txt' /etc/scalpel/scalpel.conf
85-# gif y 5000000 \x47\x49\x46\x38\x37\x61 \x00\x3b
86-# gif y 5000000 \x47\x49\x46\x38\x39\x61 \x00\x3b
87-# jpg y 5242880 \xff\xd8\xff???Exif \xff\xd9 REVERSE
88-# jpg y 5242880 \xff\xd8\xff???JFIF \xff\xd9 REVERSE
89-#
90-#
91-# PNG
92-# png y 20000000 \x50\x4e\x47? \xff\xfc\xfd\xfe
93-#
--
136-#
137-#

(kali@kali)-[~/Lab3_Data_Carving]
$ grep -n -A5 -B2 -E 'jpg|doc|txt' /etc/scalpel/scalpel.conf
85-# gif y 5000000 \x47\x49\x46\x38\x37\x61 \x00\x3b
86-# gif y 5000000 \x47\x49\x46\x38\x39\x61 \x00\x3b
87-# jpg y 5242880 \xff\xd8\xff???Exif \xff\xd9 REVERSE
88-# jpg y 5242880 \xff\xd8\xff???JFIF \xff\xd9 REVERSE
89-#
90-#
91-# PNG
92-# png y 20000000 \x50\x4e\x47? \xff\xfc\xfd\xfe
93-#
--
136-#
137-#
138-# Word documents
139-#
140-#
141-# doc y 10000000 \xd0\xcf\x11\xe0\xa1\xb1\xa1\xe1\x00\x00 \xd0\xcf\x11\xe0\xa1\xb1\xa1\xe1\x00\x00 NEXT
142-# doc y 10000000 \xd0\xcf\x11\xe0\xa1\xb1
143-#
144-# Outlook files
145-# pst y 500000000 \x21\x42\x4e\xa5\x6f\xb5\xa6
146-# ost y 500000000 \x21\x42\x44\x4e
147-#
--
194-# pgp y 100000 \xa6\x00
195-# (there should be a trailer for this ...)
196-# txt y 100000 -----BEGIN\040PGP
197-#
198-#
199-#
200-# RPM (Linux package format)
201-#
```

Backing Up the Configuration

Before modifying the Scalpel configuration, a backup of the original configuration file was created using `sudo cp`. The original `/etc/scalpel/scalpel.conf` file was copied to `/etc/scalpel/scalpel.conf.bak`. Creating this backup preserves the original configuration and provides a means of restoring the tool to its previous state if required. This is consistent with maintaining a controlled and reproducible forensic examination environment.

```
(kali@kali)-[~/Lab3_Data_Carving]
$ sudo sed -i '87,88s/^#//' /etc/scalpel/scalpel.conf
sudo sed -i '141,142s/^#//' /etc/scalpel/scalpel.conf

(kali@kali)-[~/Lab3_Data_Carving]
$ grep -n -E '^([[:space:]]*)(jpg|doc|txt|pdf|png|gif|)' /etc/scalpel/scalpel.conf
87:  jpg y 5242880 \xff\xd8\xff???Exif \xff\xd9 REVERSE
88:  jpg y 5242880 \xff\xd8\xff???JFIF \xff\xd9 REVERSE
141: doc y 10000000 \xd0\xcf\x11\xe0\xa1\xb1\xa1\xe1\x00\x00 \xd0\xcf\x11\xe0\xa1\xb1\xa1\xe1\x00\x00 NEXT
142: doc y 10000000 \xd0\xcf\x11\xe0\xa1\xb1

(kali@kali)-[~/Lab3_Data_Carving]
$ mkdir -p ~/Lab3_Data_Carving/scalpel_output
```

Enabling JPEG and DOC Signatures

The Scalpel configuration was modified to enable the file signatures relevant to the investigation. The two JPEG signatures were enabled to allow Scalpel to identify JPEG

files using EXIF or JFIF headers, while the Microsoft Word document signatures were enabled to identify Compound File Binary Format documents. The grep command was then used to verify that the four relevant configuration entries were no longer commented out. The displayed configuration confirms that Scalpel was prepared to search for JPEG and Microsoft Word files during the subsequent carving process.

```
(kali@kali)-[~/Lab3_Data_Carving]
$ mkdir -p ~/Lab3_Data_Carving/scalpel_output

(kali@kali)-[~/Lab3_Data_Carving]
$ sudo scalpel -c /etc/scalpel/scalpel.conf \
-i ~/Lab3_Data_Carving/original_evidence/Ch01InChap01.dd \
-o ~/Lab3_Data_Carving/scalpel_output \
| tee ~/Lab3_Data_Carving/tsk_analysis/scalpel_run.txt

Opening target "/home/kali/Lab3_Data_Carving/4',k0nIHC"

ERROR: Couldn't open input file: 4',k0nIHC -- No such file or directory
Scalpel was unable to open the image file: 4',k0nIHC
Skipping...

Opening target "/home/kali/Lab3_Data_Carving/4't"4'
ERROR: Couldn't open input file: 4't"4' -- No such file or directory
Scalpel was unable to open the image file: 4't"4'
Skipping...

Opening target "/home/kali/Lab3_Data_Carving/6;|4'f4'4'4'4'6*|4'
-- No such file or directory: 6;|4'f4'4'4'4'6*|4'
Scalpel was unable to open the image file: 6;|4'f4'4'4'4'6*|4'
Skipping...

Opening target "/home/kali/Lab3_Data_Carving/Non-System disk or disk error
```

Scalpel File Carving

Scalpel was used as an independent file-carving tool to examine the Ch01InChap01.dd forensic image for files that could be identified from their file signatures. The Scalpel configuration was modified to enable JPEG and Microsoft Word (.doc) signatures, while ordinary text files were not enabled because plain-text files do not have a consistent file header and footer suitable for reliable signature-based carving. The forensic image was then processed using Scalpel, which reported zero JPEG files and ten DOC files carved from the image.

```
File Actions Edit View Help

(kali@kali)-[~/Lab3_Data_Carving]
$ sudo scalpel \
-c /etc/scalpel/scalpel.conf \
-o ~/Lab3_Data_Carving/scalpel_output \
~/Lab3_Data_Carving/original_evidence/Ch01InChap01.dd
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/Lab3_Data_Carving/original_evidence/Ch01InChap01.dd"

Image file pass 1/2.
/home/kali/Lab3_Data_Carving/original_evidence/Ch01InChap01.dd: 100.0% |*****| 1.4 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" -> 0 files
jpg with header "\xff\xd8\xff\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" -> 0 files
doc with header "\xd0\xcf\x11\xe0\x1a\x1a\x1a\x00\x00" and footer "\xd0\xcf\x11\xe0\x1a\x1a\x1a\x00\x00" -> 5 files
doc with header "\xd0\xcf\x11\xe0\x1a\x1a" and footer "" -> 5 files
Carving files from image.
Image file pass 2/2.
/home/kali/Lab3_Data_Carving/original_evidence/Ch01InChap01.dd: 100.0% |*****| 1.4 MB 00:00 E
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 10, elapsed = 0 seconds.
```

The ten DOC results were distributed across two Scalpel output directories corresponding to the two enabled DOC signatures. The recovered files ranged considerably in size, from 6,144 bytes to approximately 1.37 MB. Several of the larger files were significantly larger than the known deleted Microsoft Word documents recovered using Sleuth Kit, indicating that signature-based carving produced additional candidates and possible false positives. One Scalpel result was 24,576 bytes, which corresponds exactly to the size of the Billing_Letter.doc file previously recovered using Sleuth Kit. However, file size alone was not considered sufficient to establish that the files were identical; further file-type and hash comparisons were required.

```
(kali@kali)-[~/Lab3_Data_Carving]
$ find ~/Lab3_Data_Carving/scalpel_output -type f -printf '%p %s bytes\n'
/home/kali/Lab3_Data_Carving/scalpel_output/audit.txt 1180 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000006.doc 1362908 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000007.doc 1353217 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000005.doc 1369052 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000008.doc 1328641 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000009.doc 1314305 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000000.doc 6144 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000002.doc 24576 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000001.doc 9691 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000004.doc 1314305 bytes
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000003.doc 14336 bytes
```

The Scalpel results demonstrate the difference between filesystem-based recovery and signature-based carving. Sleuth Kit was able to use the FAT12 filesystem metadata and deleted directory entries to identify and recover specific deleted files, whereas Scalpel searched for matching file signatures without relying on the filesystem's directory structure. Consequently, Scalpel generated multiple DOC candidates, including oversized results, while failing to recover the JPEG and ordinary TXT files. This demonstrates that signature-based carving can be useful for recovering files when filesystem metadata is unavailable or damaged, but its results require validation to distinguish genuine evidence from false positives or incomplete/over-carved files.

In summary, Scalpel reported:

- 0 JPEGs carved
- 10 DOC candidates carved
- 5 DOC candidates using the first DOC rule
- 5 using the second DOC rule
- Some are clearly very large false positives, ranging from ~1.3–1.4 MB
- Three of the smaller results are much more plausible:
 - 6,144 bytes
 - 24,576 bytes

- 9,691 bytes
- plus a 14,336-byte candidate

The 24,576-byte file is especially interesting because the Sleuth Kit recovery of Billing_Letter.doc was also 24 KB. The Sleuth Kit uses filesystem metadata and directory information, whereas Scalpel relies primarily on file signatures.

Because the FAT12 image contains fragmented/deleted data and repeated OLE/Word signatures, Scalpel can produce oversized or duplicate/false-positive carvings.

```
(kali@kali) - [~/Lab3_Data_Carving]
$ file ~/Lab3_Data_Carving/scalpel_output/doc-2-0/*.doc \
~/Lab3_Data_Carving/scalpel_output/doc-3-0/*.doc
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000000.doc: Composite Document File V2 Document, Can't read SSAT
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000001.doc: Composite Document File V2 Document, Cannot read section info
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000002.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000003.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec 9 14:52:18 2005, Security: 0
/home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000004.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 02 November 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 02:00, Create Time/Date: Sat Nov 23 03:12:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 58, Number of Characters: 336, Security: 0
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000005.doc: Composite Document File V2 Document, Can't read SSAT
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000006.doc: Composite Document File V2 Document, Cannot read section info
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000007.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000008.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec 9 14:52:18 2005, Security: 0
/home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000009.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 02 November 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 02:00, Create Time/Date: Sat Nov 23 03:12:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 58, Number of Characters: 336, Security: 0
```

From the screenshot above, the following can be confirm,

Scalpel file	Interpretation
doc-2-0/00000002.doc	Valid Word document - Billing Letter
doc-2-0/00000004.doc	Valid Word document - Regrets
doc-3-0/00000007.doc	Duplicate of Billing Letter
doc-3-0/00000009.doc	Duplicate of Regrets
00000000/05	Invalid/incomplete DOC carving
00000001/06	Invalid/incomplete DOC carving
00000003/08	Actually identifies as a Microsoft Excel-created OLE document, corresponding to the filesystem's Income.xls

Scalpel file	Interpretation
JPEG	None carved

The metadata of Scalpel's valid files exactly matches the metadata we previously obtained from our Sleuth Kit recoveries.

For example, 00000002.doc has:

- Title: 13 October 2003
- Author: Amelia Phillips
- 118 words
- 678 characters
- Microsoft Word 9.0
- Same creation/last-saved timestamps

And 00000004.doc similarly matches the metadata of Regrets.doc.

So now let's prove whether the actual recovered bytes are identical.

The Scalpel files are not byte-for-byte identical to the Sleuth Kit recoveries.

Hash comparison

```
(kali@kali) - [~/Lab3_Data_Carving]
$ md5sum \
~/Lab3_Data_Carving/recovered/Billing_Letter.doc \
~/Lab3_Data_Carving/scalpel_output/doc-2-0/00000002.doc \
~/Lab3_Data_Carving/scalpel_output/doc-3-0/00000007.doc
9fe241d0dde27e83442010b3eee5ad32 /home/kali/Lab3_Data_Carving/recovered/Billing_Letter.doc
3521959b966ff72bbcd833519d26efdc /home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000002.doc
9e08b149e15a71e39e029f3b7a3e73b7 /home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000007.doc

(kali@kali) - [~/Lab3_Data_Carving]
$ sha256sum \
~/Lab3_Data_Carving/recovered/Billing_Letter.doc \
~/Lab3_Data_Carving/scalpel_output/doc-2-0/00000002.doc \
~/Lab3_Data_Carving/scalpel_output/doc-3-0/00000007.doc
7eefeb4187bb551f366da33bcab5b7224f905bb98f37797c28f183f72aec2e99 /home/kali/Lab3_Data_Carving/recovered/Billing_Letter.doc
ed7dbbd47518ec93da1bb52acd57f8ebcfb22b2685cb327170afa54d2ebb4076 /home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000002.doc
b934a276d8a70a5461ba159f0ec0f068534ceedf49ea0af0b21e36036272bd25 /home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000007.doc
```

Billing Letter

Recovery	MD5	SHA-256
Sleuth Kit	9fe241d0dde27e83442010b3eee5ad32	7eefeb4187bb551f366da33bcab5b7224f905bb98f37797c28f183f72aec2e99

Recover y	MD5	SHA-256
Scalpel 0000000 2.doc	3521959b966ff72bbed83 3519d26efdc	ed7dbbd47518ec93da1bb52acd57f8ebcfb22b268 5cb327170afa54d2ebb4076
Scalpel 0000000 7.doc	9e08b149e15a71e39e02 9f3b7a3e73b7	b934a276d8a70a5461ba159f0ec0f068534ceedf4 9ea0af0b21e36036272bd25

Regrets

```
(kali@kali) ~/Lab3_Data_Carving
$ md5sum \
~/Lab3_Data_Carving/recovered/Regrets.doc \
~/Lab3_Data_Carving/scalpel_output/doc-2-0/00000004.doc \
~/Lab3_Data_Carving/scalpel_output/doc-3-0/00000009.doc
ebcfbf22bdf81a60f6a16709d30c1dad /home/kali/Lab3_Data_Carving/recovered/Regrets.doc
bb47d834b7e43a4bd12a95eb6184a562 /home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000004.doc
bb47d834b7e43a4bd12a95eb6184a562 /home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000009.doc

(kali@kali) ~/Lab3_Data_Carving
$ sha256sum \
~/Lab3_Data_Carving/recovered/Regrets.doc \
~/Lab3_Data_Carving/scalpel_output/doc-2-0/00000004.doc \
~/Lab3_Data_Carving/scalpel_output/doc-3-0/00000009.doc
109048540a0a1a452f9346522a325219b471eae41b85a6298890e94ae750b965 /home/kali/Lab3_Data_Carving/recovered/Regrets.doc
279d2cff80f9c2a09631f1375c3058ddf0b2aca728b16342936acf2ed66d8029 /home/kali/Lab3_Data_Carving/scalpel_output/doc-2-0/00000004.doc
279d2cff80f9c2a09631f1375c3058ddf0b2aca728b16342936acf2ed66d8029 /home/kali/Lab3_Data_Carving/scalpel_output/doc-3-0/00000009.doc
```

Recover y	MD5	SHA-256
Sleuth Kit	ebcfbf22bdf81a60f6a167 09d30c1dad	109048540a0a1a452f9346522a325219b471eae4 1b85a6298890e94ae750b965
Scalpel 0000000 4.doc	bb47d834b7e43a4bd12a 95eb6184a562	279d2cff80f9c2a09631f1375c3058ddf0b2aca728b 16342936acf2ed66d8029
Scalpel 0000000 9.doc	bb47d834b7e43a4bd12a 95eb6184a562	279d2cff80f9c2a09631f1375c3058ddf0b2aca728b 16342936acf2ed66d8029

We now have evidence that Sleuth Kit:

- Recovered the deleted files using FAT12 filesystem metadata.
- Produced complete, valid files.
- Preserved the original recovered file structure/data sufficiently for consistent hashing.

Scalpel

- Found the same underlying Word-document content/signatures.
- Produced valid-looking Word documents for some candidates.
- Produced different byte sequences, despite matching document metadata.
- Produced duplicates and false positives.
- Produced several oversized/incomplete candidates.

Scalpel Carving: Billing Letter:

This screenshot shows the content recovered by Scalpel from the forensic image. The carved document contains identifiable billing-letter content and Word metadata associated with the Billing Letter. It also shows additional text recovered beyond the original document boundary, demonstrating the possibility of over-carving when using signature-based file carving.

```

kali@kali:~/Lab3_Data_Carving$ strings -a ~/Lab3_Data_Carving/scalpel_output/doc-2-0/00000002.doc | head -80
bjbJU
13 October 2005
Laura Roper
48 Mockingbird Lane
Seattle, WA 98119
Dear Laura,
Per our conversation, you want to register a domain with the name of
HYPERLINK http://www.lauras_stuff.com
www.lauras_stuff.com
. You have already chosen IT Connection Servers to host your website.
I have been in contact with them to get the necessary information. My fee for the registration process is $500. Please send a check or money order to the address listed below.
George Montgomery
3467 Main Street
Bellevue, WA 98008
If you fail to do so within 30 days of receipt of this letter, the domain ownership will revert to me and I will sell it to the highest bidder.
Please call me at (206) 555-1212 or email me at
HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
Regards,
George Montgomery
13 October 2003
Amelia Phillips
Normal.dot
Amelia Phillips
Microsoft Word 9.0
Starships CMS
13 October 2003
Title
_PID_HLINKS
Microsoft Word Document
MSWordDoc
Word.Document.8
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.
ftp.acmeserver.com
login id: laura.roper
password: 342rr0iu9
Thank you for your business

```

Scalpel Carving: Regrets Letter:

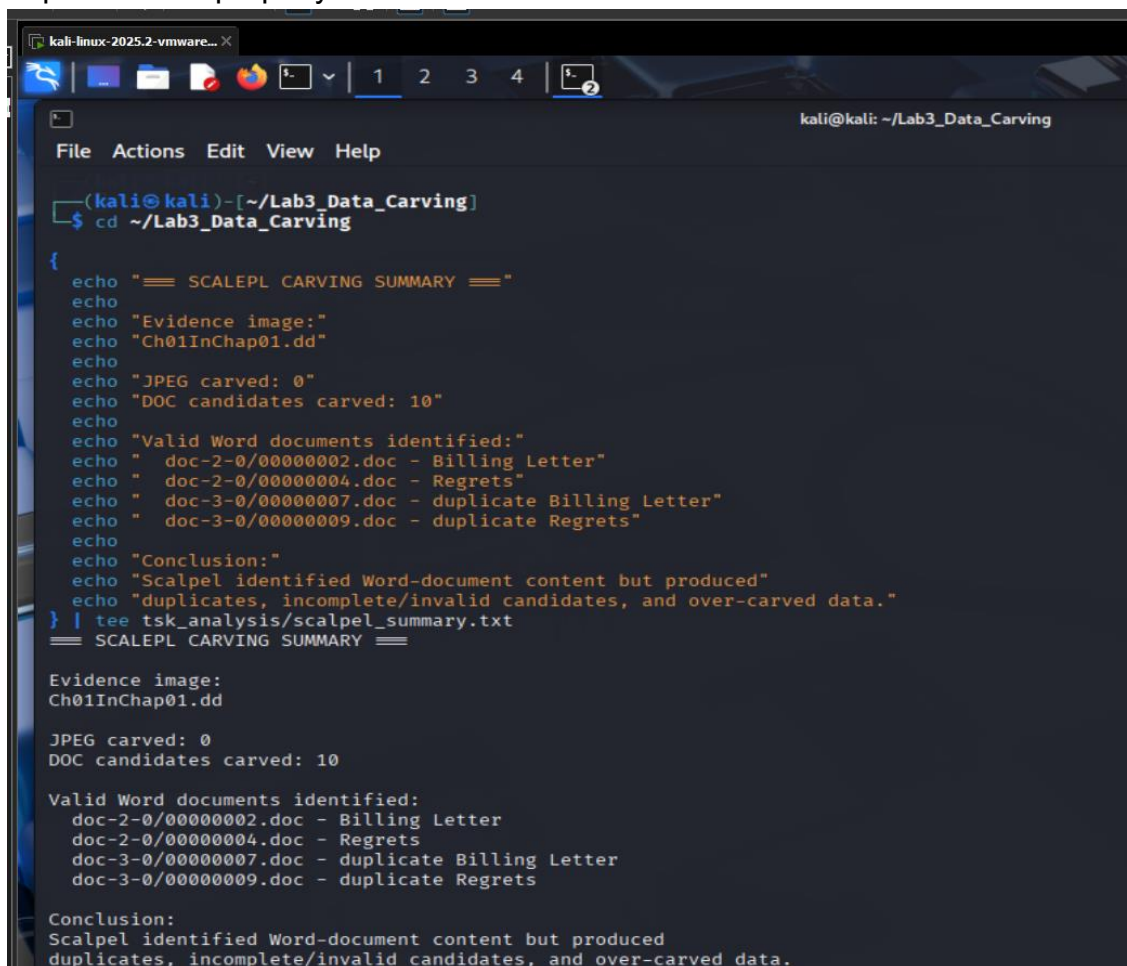
This screenshot shows the content recovered from the second Scalpel-carved Word document. The output contains the expected Regrets Letter text together with associated Microsoft Word metadata, confirming that Scalpel successfully identified the deleted document content.

```

(kali@kali)-[~/Lab3_Data_Carving]
$ strings -a ~/Lab3_Data_Carving/scalpel_output/doc-2-0/00000004.doc | head -80
bjbjU
02 November 2005
Randall Watson
89 Darnell Street
Des Moines, WA 98000
Dear Mr. Watson,
I regret to inform you that all five versions of the domain name that you chose have already been purchased. If you have other variations, please let me
ow by calling me at (206) 555-1212 or email me at
HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
Regards,
George Montgomery
02 November 2003
Amelia Phillips
Normal.dot
Amelia Phillips
Microsoft Word 9.0
Starships CMS
02 November 2003
Title
PID_HLINKS
Microsoft Word Document
MSWordDoc
Word.Document.8

```

The screenshots below document the execution and verification of the Scalpel carving process against the Ch01InChap01.dd forensic image. The results show that Scalpel carved 10 DOC candidates but did not recover any JPEG files. Further examination of the carved files identified valid Word documents corresponding to the previously recovered Billing Letter and Regrets files, alongside duplicate, incomplete, and over-carved results. This demonstrates the ability of signature-based carving to recover deleted document content while also highlighting the possibility of false positives and duplicate or improperly bounded files.



```

kali@kali: ~/Lab3_Data_Carving
File Actions Edit View Help

(kali@kali)-[~/Lab3_Data_Carving]
$ cd ~/Lab3_Data_Carving

{
echo "=== SCALEPL CARVING SUMMARY ==="
echo
echo "Evidence image:"
echo "Ch01InChap01.dd"
echo
echo "JPEG carved: 0"
echo "DOC candidates carved: 10"
echo
echo "Valid Word documents identified:"
echo "  doc-2-0/00000002.doc - Billing Letter"
echo "  doc-2-0/00000004.doc - Regrets"
echo "  doc-3-0/00000007.doc - duplicate Billing Letter"
echo "  doc-3-0/00000009.doc - duplicate Regrets"
echo
echo "Conclusion:"
echo "Scalpel identified Word-document content but produced"
echo "duplicates, incomplete/invalid candidates, and over-carved data."
} | tee tsk_analysis/scalpel_summary.txt
=== SCALEPL CARVING SUMMARY ===

Evidence image:
Ch01InChap01.dd

JPEG carved: 0
DOC candidates carved: 10

Valid Word documents identified:
  doc-2-0/00000002.doc - Billing Letter
  doc-2-0/00000004.doc - Regrets
  doc-3-0/00000007.doc - duplicate Billing Letter
  doc-3-0/00000009.doc - duplicate Regrets

Conclusion:
Scalpel identified Word-document content but produced
duplicates, incomplete/invalid candidates, and over-carved data.

```

The output confirms that Scalpel successfully identified recoverable Microsoft Word document structures within the forensic image. However, the presence of multiple candidates for the same documents and invalid or incomplete carvings indicates that the results required further validation using file identification, content inspection, and cryptographic hashing.

```
(kali㉿kali)-[~/Lab3_Data_Carving]
$ cat tsk_analysis/scalpel_summary.txt
== SCALEPL CARVING SUMMARY ==

Evidence image:
Ch01InChap01.dd

JPEG carved: 0
DOC candidates carved: 10

Valid Word documents identified:
doc-2-0/00000002.doc - Billing Letter
doc-2-0/00000004.doc - Regrets
doc-3-0/00000007.doc - duplicate Billing Letter
doc-3-0/00000009.doc - duplicate Regrets

Conclusion:
Scalpel identified Word-document content but produced
duplicates, incomplete/invalid candidates, and over-carved data.
```

Evidence Acquisition and Preservation

The forensic investigation began by identifying and preserving the supplied evidence files before conducting any analysis. The evidence consisted of a JPEG image (J_ub_law.jpg), a 120 MB compressed archive (120M.7z), and a forensic disk image (Ch01InChap01.dd). The original evidence was stored separately from the working copies and recovered files to maintain evidence integrity. Cryptographic hashes were calculated using MD5 and SHA-256 to provide a baseline for subsequent verification. The SHA-256 hash of Ch01InChap01.dd was 3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122, while its MD5 hash was a117773bcf1fc88ec0ab8e0a349fbbcb. These hashes can be used to demonstrate that the original evidence image remained unchanged during the examination.

JPEG File Analysis Using XXD

The JPEG file J_ub_law.jpg was examined at the hexadecimal level using the xxd utility. Examination of the file header confirmed the presence of the JPEG Start of Image (SOI) signature, represented by the hexadecimal sequence FF D8, while the file footer contained the JPEG End of Image (EOI) marker FF D9. These signatures confirmed that the file had a valid JPEG structure. A working copy of the image was subsequently examined rather than modifying the original evidence. The reconstructed JPEG was compared against the original using MD5 and SHA-256 hashing, and both hashes

matched the original values. This confirmed that the reconstruction process did not alter the evidence.

JPEG Metadata and String Analysis

Further examination of the JPEG was performed using strings and exiftool to identify embedded textual information and metadata. The file contained EXIF information identifying the camera manufacturer as NIKON CORPORATION and the camera model as NIKON D4. String analysis was also used to identify human-readable information embedded within the file. This demonstrated how metadata and residual strings can provide useful contextual information during forensic examination, even when the primary purpose of the investigation is file carving.

Binwalk Analysis of the JPEG

Binwalk was used to examine the JPEG for embedded file signatures and other identifiable structures. The analysis identified JPEG image data containing EXIF information and a TIFF structure beginning at offset 0x0C. Binwalk also identified a copyright string associated with Adobe Systems Incorporated at hexadecimal offset 0x68DC. No separate embedded file was identified through the Binwalk scan. The result demonstrated that the JPEG contained identifiable internal structures and metadata but did not reveal an additional independently embedded file through signature scanning.

Filesystem Identification Using Sleuth Kit

The forensic image Ch01InChap01.dd was examined using the Sleuth Kit utilities `img_stat`, `mmls`, and `fsstat`. `img_stat` identified the evidence as a raw disk image with a size of 1,474,560 bytes and a sector size of 512 bytes. The `mmls` command produced no partition-table output, indicating that the image did not contain a conventional partition table and that the filesystem began directly at the start of the image. Further analysis with `fsstat` identified the filesystem as FAT12. The filesystem contained a 512-byte sector size and a 512-byte cluster size, with the data area beginning at sector 19. This filesystem-level information was important because it allowed deleted directory entries and their associated data clusters to be examined.

Deleted File Identification

The `fls` utility was used to examine the FAT12 filesystem recursively and identify existing and deleted directory entries. The analysis identified several files, including Client Info.mdb, Income.xls, and four entries marked as deleted: Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. The asterisk associated with these entries indicated that they had been deleted but that their directory information was still available within the filesystem. This demonstrated the value of filesystem-aware forensic analysis because the deleted filenames and their filesystem metadata could still be identified even after deletion.

Recovery of Deleted Files Using ICAT

The deleted files identified through fls were recovered using the Sleuth Kit icat utility with the recovery option. Four deleted files were successfully recovered: Billing_Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. The recovered Word documents were identified as Microsoft Compound Document Files, while the two text files were identified as ASCII text. The recovered Billing_Letter.doc was approximately 24 KB in size, while Regrets.doc was approximately 23 KB. The successful recovery demonstrated that deletion from a FAT12 filesystem does not necessarily result in immediate destruction of the underlying file data.

Analysis of Recovered Evidence

The recovered files were examined using the file and strings utilities. The analysis of Billing_Letter.doc revealed information concerning a domain registration transaction, including the name of the intended domain owner, the registration fee, and contact information associated with the sender. The Regrets.doc file contained a letter informing a recipient that previously selected domain names had already been purchased. The letter1.txt file contained a message requesting a meeting on the 18th of August, while confirmation.txt contained information relating to domain registration and FTP access. Because the latter file contained authentication information, the credential itself was treated as sensitive evidence and was not reproduced in the public report.

An interesting metadata discrepancy was also observed in the Word documents. The visible document content referenced dates in 2005, whereas the embedded Word metadata contained corresponding titles dated 2003. This discrepancy was documented as an observation rather than interpreted as proof of manipulation, since metadata and visible document content can represent different events or editing states.

Hash Verification of Recovered Files

Cryptographic hashes were calculated for each recovered file to establish their identities and support evidence verification. The MD5 and SHA-256 values were recorded for Billing_Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. For example, the MD5 hash of Billing_Letter.doc was 9fe241d0dde27e83442010b3eee5ad32, while its SHA-256 hash was 7eefeb4187bb551f366da33bcab5b7224f905bb98f37797c28f183f72aec2e99. Recording these values provides a reproducible means of demonstrating that the recovered files remain unchanged after recovery and analysis.

Binwalk Analysis of the Forensic Image

Binwalk was also executed against Ch01InChap01.dd to identify additional embedded file signatures that might not have been identified through filesystem analysis. The scan returned no recognizable embedded signatures. This result did not indicate that the image contained no recoverable files; rather, it demonstrated that Binwalk's signature-based approach did not identify additional structures beyond those already detected through

filesystem-aware analysis. The finding reinforced the importance of using multiple forensic techniques when examining digital evidence.

File Carving Using Scalpel

Scalpel was used as an independent file-carving technique to examine Ch01InChap01.dd based on file headers and footers rather than filesystem metadata. JPEG and Microsoft Word document signatures were enabled in the Scalpel configuration. The carving process identified 10 DOC candidates, while no JPEG files were carved. Examination using the file utility showed that several candidates were incomplete or invalid, while others were valid Compound Document Files. Two candidates corresponded to the previously recovered Word documents: 00000002.doc corresponded to the Billing Letter and 00000004.doc corresponded to the Regrets document. Duplicate candidates were also produced, demonstrating the limitations of signature-based carving.

Comparison of Sleuth Kit and Scalpel Results

A comparison between the Sleuth Kit and Scalpel results demonstrated that the two techniques produced complementary findings. Scalpel successfully identified Word-document content corresponding to the Billing Letter and Regrets documents, and the document metadata and substantive text were consistent with the files recovered using Sleuth Kit. However, the MD5 and SHA-256 hashes of the Scalpel-carved files differed from those of the Sleuth Kit recoveries. This indicates that the files were not byte-for-byte identical. Scalpel also produced incomplete candidates, duplicate files, and over-carved data. In one case, the Scalpel carving of the Billing Letter contained additional textual data belonging to another recovered item. These results demonstrate that signature-based carving can recover useful evidence but may also produce false positives or incorrect file boundaries.

Overall Findings

The investigation demonstrated the effectiveness of combining hexadecimal examination, metadata analysis, filesystem analysis, signature analysis, and file carving during a digital forensic examination. XXD and metadata analysis provided structural and contextual information about the JPEG evidence, while Sleuth Kit successfully identified the FAT12 filesystem and recovered deleted files using filesystem metadata. Binwalk provided additional signature-based examination, while Scalpel independently demonstrated the presence of recoverable Word-document content. The comparison showed that filesystem-aware recovery produced more forensically meaningful results for the deleted files, while Scalpel served as useful corroborating evidence but generated additional false positives and carving artifacts.

Conclusion

The practical investigation successfully demonstrated the process of preserving, examining, recovering, and validating digital evidence. Four deleted files were

successfully recovered from the FAT12 forensic image using Sleuth Kit, and their contents and metadata were examined. Independent Scalpel carving corroborated the presence of two of the recovered Word documents but also demonstrated the limitations of signature-based recovery through duplicate, incomplete, and over-carved results. Hash verification was used throughout the investigation to support evidence integrity. Overall, the exercise demonstrates that reliable digital forensic analysis benefits from the combination of filesystem-aware recovery and file-carving techniques rather than relying on a single recovery method.